Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Hillcrest Dairy LLC

Physical address of dairy:

Number and Street	City	County	Zip Code
1901 N Hayden RD	Le Grand	Merced	95333

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 06/12/2002

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X053-X100-X043-XXXX X053-X100-X073-XXXX

B. OPERATORS

Hoekstra, Edward

Operator name: Hoekstra, Edward

Telephone no.: Landline: ; Cellular: (209) 535-8591

Mailing Address Number and Street	City	State	Zip Code
1901 N Hayden RD	Le Grand	CA	95333

This operator is responsible for paying permit fees.

C. OWNERS

Hoekstra, Edward

Legal owner name: Hoekstra, Edward

Telephone no.: Landline: ; Cellular: (209) 535-8591

Mailing Address Number and Street	City	State	Zip Code
1901 N Hayden RD	Le Grand	CA	95333

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	750	1,400	0	0	0
Number under roof	4,000	0	0	500	1,400	0
Maximum number	4,000	750	1,400	500	1,400	0
Average number	4,000	750	1,400	500	1,400	0
Avg live weight (lbs)	1,400	1,500	900	700		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 138,422.69 *tons per reporting period*

Total nitrogen from manure: 1,742,521.08 *lbs per reporting period*

After ammonia losses (30% loss applied): 1,219,764.76 *lbs per reporting period*

Total phosphorus from manure: 283,008.90 *lbs per reporting period*

Total potassium from manure: 771,292.91 *lbs per reporting period*

Total salt from manure: 2,055,862.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 163,336,320 *gallons*

Total nitrogen generated: 288,459.95 *lbs*

Total phosphorus generated: 52,358.59 *lbs*

Total potassium generated: 536,743.09 *lbs*

Total salt generated: 2,690,142.83 *lbs*

	163,336,320	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	163,336,320	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
10990 Bear Creek	Surface water
D135	Surface water
MID Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/01/2023	Solid commercial fertilizer *10-0-8*	267.00 *ton*	Dry-weight	0.1	10.000000	0.000000	8.000000	0.000000
06/20/2023	Solid commercial fertilizer *UN32*	52.00 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	86,593.32	0.00	42,677.28	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	86,593.32	0.00	42,677.28	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	Corral solids	1,000.00 *ton*	Dry-weight	33.4		13,200.00	9,000.00	34,000.00		0.00
09/06/2023	Corral solids	750.00 *ton*	Dry-weight	33.4		13,200.00	9,000.00	34,000.00		0.00
09/06/2023	Corral solids	500.00 *ton*	Dry-weight	33.4		13,200.00	9,000.00	34,000.00		0.00
09/06/2023	Corral solids	750.00 *ton*	Dry-weight	33.4		13,200.00	9,000.00	34,000.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	52,747.20	35,964.00	135,864.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	52,747.20	35,964.00	135,864.00	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	59	59	1	process wastewater	X053-X080-X045-XXXX
12	93	93	2	process wastewater	X053-X100-X043-XXXX
13	79	79	2	process wastewater	X053-X100-X043-XXXX
14	74	74	2	process wastewater	X053-X100-X043-XXXX
15	77	77	2	process wastewater	X053-X100-X043-XXXX
16	50	50	2	process wastewater	X053-X100-X044-XXXX
17	28	28	1	process wastewater	X053-X110-X006-XXXX
18	49	49	2	process wastewater	X053-X150-X033-XXXX
2	9	9	1	process wastewater	X053-X080-X045-XXXX
20 N	37	37	1	process wastewater	X053-X150-X033-XXXX
21	15	15	1	none	X053-X150-X033-XXXX
22	115	115	1	process wastewater	X053-X150-X006-XXXX
24 E	81	81	2	process wastewater	X037-X040-X004-XXXX X053-X150-X032-XXXX
25	92	92	1	none	X053-X150-X006-XXXX
26	76	76	2	manure	X053-X110-X006-XXXX
3	33	33	1	process wastewater	X053-X080-X045-XXXX
31	78	78	0	none	X053-X100-X065-XXXX
4	76	76	2	process wastewater	X053-X080-X045-XXXX
5	36	36	1	process wastewater	X053-X100-X069-XXXX
6	57	57	2	process wastewater	X053-X100-X069-XXXX
7	13	13	1	process wastewater	X053-X100-X073-XXXX
8	27	27	2	process wastewater	X053-X100-X073-XXXX
Hill Field	40	40	0	none	X053-X100-X013-XXXX
Pistachios	464	464	1	none	X037-X040-X003-XXXX X053-X100-X042-XXXX X053-X100-X047-XXXX X053-X100-X065-XXXX X053-X150-X033-XXXX
Totals for areas that were used for application	1,069	1,069	30		
Totals for areas that were not used for application	689	689	3		
Land application area totals	1,758	1,758	33		

B. CROPS AND HARVESTS

1

Field name: 1

04/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 59 Plant date: 04/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/08/2023	2,073.00 *ton*	As-is		68.0	4,060.00	760.00	5,550.00		5.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	35.14	285.30	53.41	390.01	1,158.07

12

Field name: 12

04/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 92 Plant date: 04/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/11/2023	2,265.00 *ton*	As-is		65.0	3,360.00	780.00	5,350.00		4.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	24.62	165.44	38.41	263.43	737.60

08/15/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 46 Plant date: 08/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/26/2023	417.00 *ton*	As-is		70.6	5,880.00	900.00	9,440.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	66.00	10.20	72.00	0.00
Total actual harvest content	9.07	106.61	16.32	171.15	565.02

13

Field name: 13

04/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 79 Plant date: 04/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/12/2023	2,873.00 *ton*	As-is		70.0	2,820.00	540.00	3,650.00		5.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	36.37	205.11	39.28	265.48	1,130.29

08/16/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 79 Plant date: 08/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	313.00 *ton*	As-is		54.2	9,020.00	1,440.00	14,900.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	66.00	10.20	72.00	0.00
Total actual harvest content	3.96	71.47	11.41	118.07	395.58

14

Field name: 14

11/06/2022: Wheat Hay

Crop: Wheat Hay Acres planted: 74 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	290.00 *ton*	As-is		10.0	16,700.00	3,780.00	21,500.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	132.00	20.40	99.60	0.00
Total actual harvest content	3.92	130.89	29.63	168.51	853.54

14

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 74 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	2,193.00 *ton*	As-is		70.0	2,760.00	5,000.00	4,220.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	29.64	163.59	296.35	250.12	1,298.02

15

Field name: 15

11/09/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 77 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	1,096.00 *ton*	As-is		68.8	5,470.00	1,090.00	7,360.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	14.23	155.72	31.03	209.52	870.42

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,814.00 *ton*	As-is		64.0	3,310.00	1,160.00	4,380.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	23.56	155.96	54.66	206.37	1,102.54

16

Field name: 16

11/08/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 50 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	796.00 *ton*	As-is		71.2	5,710.00	890.00	6,490.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	15.92	181.81	28.34	206.64	962.84

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,262.00 *ton*	As-is		63.6	3,390.00	1,130.00	6,450.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	25.24	171.13	57.04	325.60	1,194.36

17

Field name: 17

04/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 04/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/07/2023	816.00 *ton*	As-is		67.8	3,800.00	700.00	4,780.00		6.26

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	29.14	221.49	40.80	278.61	1,174.88

18

Field name: 18

11/03/2022: WheatHay

Crop: WheatHay Acres planted: 49 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	154.00 *ton*	As-is		10.0	11,700.00	2,250.00	11,900.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	132.00	20.40	99.60	0.00
Total actual harvest content	3.14	73.54	14.14	74.80	492.17

07/29/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 49 Plant date: 07/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/03/2023	369.00 *ton*	As-is		62.4	4,140.00	1,020.00	9,890.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	66.00	10.20	72.00	0.00
Total actual harvest content	7.53	62.35	15.36	148.96	583.29

2

Field name: 2

04/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 04/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/08/2023	328.00 *ton*	As-is		63.0	4,590.00	670.00	6,000.00		5.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	36.44	334.56	48.84	437.33	1,531.83

20 N

Field name: 20 N

04/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 04/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/10/2023	974.00 *ton*	As-is		65.6	4,270.00	930.00	5,310.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	27.83	237.66	51.76	295.54	1,110.47

21

Field name: 21

04/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 04/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/10/2023	340.00 *ton*	As-is		62.0	4,750.00	980.00	5,810.00		6.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	22.67	215.33	44.43	263.39	1,042.21

22

Field name: 22

22

04/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 115 Plant date: 04/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/11/2023	3,068.00 *ton*	As-is		65.6	3,780.00	860.00	5,700.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.68	201.69	45.89	304.13	1,046.21

24 E

Field name: 24 E

11/10/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 81 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	1,134.00 *ton*	As-is		62.0	5,610.00	1,070.00	6,410.00		9.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	14.00	157.08	29.96	179.48	1,010.80

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 81 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	2,541.00 *ton*	As-is		72.2	2,840.00	850.00	4,390.00		2.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	31.37	178.18	53.33	275.43	451.75

25

Field name: 25

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 92 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,817.00 *ton*	As-is		63.4	4,320.00	820.00	4,550.00		2.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	19.75	170.64	32.39	179.73	404.80

26

Field name: 26

11/11/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 74 Plant date: 11/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	790.00 *ton*	As-is		52.2	7,080.00	1,400.00	7,090.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	10.68	151.17	29.89	151.38	1,122.65

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 74 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/15/2023	2,056.00 *ton*	As-is		64.9	4,700.00	580.00	4,830.00		6.96

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	27.78	261.17	32.23	268.39	1,357.49

3

Field name: 3

04/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 04/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/09/2023	1,074.00 *ton*	As-is		67.8	3,900.00	780.00	5,960.00		0.76

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	32.55	253.85	50.77	387.94	159.29

4

Field name: 4

04/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 76 Plant date: 04/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/07/2023	2,146.00 *ton*	As-is		63.6	4,080.00	690.00	5,120.00		4.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	28.24	230.41	38.97	289.15	863.37

08/14/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 76 Plant date: 08/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	591.00 *ton*	As-is		70.4	6,630.00	720.00	7,830.00		8.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	66.00	10.20	72.00	0.00
Total actual harvest content	7.78	103.11	11.20	121.78	386.70

5

Field name: 5

04/17/2002: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 04/17/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/09/2023	1,263.00 *ton*	As-is		70.2	3,670.00	830.00	4,420.00		6.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	35.08	257.51	58.24	310.14	1,292.22

6

Field name: 6

04/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 57 Plant date: 04/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/09/2023	1,803.00 *ton*	As-is		66.4	4,000.00	920.00	4,330.00		5.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	31.63	253.05	58.20	273.93	1,098.96

08/14/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 57 Plant date: 08/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	401.00 *ton*	As-is		67.6	7,350.00	760.00	8,360.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	66.00	10.20	72.00	0.00
Total actual harvest content	7.04	103.42	10.69	117.63	341.91

7

Field name: 7

04/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 13 Plant date: 04/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/10/2023	332.00 *ton*	As-is		64.7	4,480.00	910.00	5,020.00		3.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	25.54	228.82	46.48	256.41	663.51

8

Field name: 8

11/02/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 9 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/03/2023	79.00 *ton*	As-is		56.4	8,130.00	1,830.00	10,400.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	176.00	27.20	132.80	0.00
Total actual harvest content	8.78	142.73	32.13	182.58	926.16

07/29/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 9 Plant date: 07/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	69.00 *ton*	As-is		61.0	7,330.00	1,280.00	12,300.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	6.00	66.00	10.20	72.00	0.00
Total actual harvest content	7.67	112.39	19.63	188.60	693.68

Pistachios

Field name: Pistachios

01/01/2022: Pistachios

Crop: Pistachios Acres planted: 464 Plant date: 01/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	900.00 *ton*	As-is		5.0	28,000.00	3,000.00	25,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.00	240.00	40.00	200.00	0.00
Total actual harvest content	1.94	108.62	11.64	96.98	0.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 04/12/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 04/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.73	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		4.73	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.68	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		3.68	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.16	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.42	0.00	0.00	0.00	8,424,000.00 *gal*
Application event totals		3.42	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	196.84	39.31	519.67	2,087.32	3,823,200.00 *gal*
MID Canal	Surface water	3.16	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		199.99	39.31	519.67	2,087.32	

1 - 04/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.53	11.80	59.74	466.85	4,460,400.00 *gal*
MID Canal	Surface water	3.68	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		60.21	11.80	59.74	466.85	

12 - 04/27/2023: Corn, silage

Field name: 12

Crop: Corn, silage

Plant date: 04/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.02	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.35	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		1.35	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.35	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		1.35	0.00	0.00	0.00	

12 - 04/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.26	22.02	291.09	1,169.20	3,339,360.00 *gal*
MID Canal	Surface water	1.01	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		111.27	22.02	291.09	1,169.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.15	16.81	222.18	892.40	2,548,800.00 *gal*
MID Canal	Surface water	0.67	0.00	0.00	0.00	2,592,000.00 *gal*
Application event totals		84.83	16.81	222.18	892.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.35	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		1.35	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.02	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

12 - 08/15/2023: Sudangrass, silage

Field name: 12

Crop: Sudangrass, silage

Plant date: 08/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

12 - 08/15/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.37	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		3.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.34	7.38	37.36	291.90	2,174,400.00 *gal*
MID Canal	Surface water	2.70	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		38.04	7.38	37.36	291.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.88	9.99	50.61	395.47	2,945,880.00 *gal*
MID Canal	Surface water	2.36	0.00	0.00	0.00	4,536,000.00 *gal*
Application event totals		50.25	9.99	50.61	395.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	5.40	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		5.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.15	0.00	0.00	0.00	6,048,000.00 *gal*
Application event totals		3.15	0.00	0.00	0.00	

13 - 04/25/2023: Corn, silage

Field name: 13

Crop: Corn, silage

Plant date: 04/25/2023

13 - 04/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.93	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		3.93	0.00	0.00	0.00	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.50	34.25	452.79	1,818.70	4,460,400.00 *gal*
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		174.25	34.25	452.79	1,818.70	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	79.66	16.63	84.20	657.93	8,416,800.00 *gal*
MID Canal	Surface water	3.93	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		83.59	16.63	84.20	657.93	

13 - 04/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

13 - 08/16/2023: Sudangrass, silage

Field name: 13

Crop: Sudangrass, silage Plant date: 08/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.14	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.14	0.00	0.00	0.00	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.16	8.59	43.50	339.94	4,348,800.00 *gal*
MID Canal	Surface water	3.14	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		44.30	8.59	43.50	339.94	

13 - 08/16/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.02	7.52	38.07	297.45	3,805,200.00 *gal*
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		38.77	7.52	38.07	297.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.75	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.75	0.00	0.00	0.00	

14 - 11/06/2022: Wheat Hay

Field name: 14

Crop: Wheat Hay

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	176.68	12.74	120.19	1,487.49	2,782,800.00 *gal*
MID Canal	Surface water	2.10	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		178.78	12.74	120.19	1,487.49	

14 - 06/09/2023: Corn, silage

14 - 06/09/2023: Corn, silage

Field name: 14

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.78	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.78	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	182.77	36.50	482.53	1,938.13	4,452,480.00 *gal*
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		186.12	36.50	482.53	1,938.13	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

14 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.94	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.94	0.00	0.00	0.00	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.94	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.94	0.00	0.00	0.00	

15 - 11/09/2022: Wheat, silage, soft dough

Field name: 15

Crop: Wheat, silage, soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.96	15.79	114.85	735.89	1,648,080.00 *gal*
MID Canal	Surface water	2.42	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		131.38	15.79	114.85	735.89	

12/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.12	5.42	51.10	632.47	1,231,200.00 *gal*
MID Canal	Surface water	1.21	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		76.33	5.42	51.10	632.47	

15 - 11/09/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.03	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		4.03	0.00	0.00	0.00	

15 - 06/19/2023: Corn, silage

Field name: 15

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.43	0.00	0.00	0.00	14,256,000.00 *gal*
Application event totals		4.43	0.00	0.00	0.00	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.63	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.63	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.22	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.22	0.00	0.00	0.00	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.63	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.63	0.00	0.00	0.00	

15 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	197.69	39.48	521.92	2,096.35	5,011,200.00 *gal*
MID Canal	Surface water	2.42	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		200.11	39.48	521.92	2,096.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.22	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.22	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	0.11	0.00	0.00	0.00	368,000.00 *gal*
Application event totals		0.11	0.00	0.00	0.00	

16 - 11/08/2022: Wheat, silage, soft dough

Field name: 16

Crop: Wheat, silage, soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.59	7.54	71.15	880.60	1,113,120.00 *gal*
MID Canal	Surface water	2.48	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		107.08	7.54	71.15	880.60	

16 - 11/08/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.48	17.47	230.96	927.70	1,440,000.00 *gal*
MID Canal	Surface water	2.48	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		89.97	17.47	230.96	927.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.95	3.98	52.66	211.51	328,320.00 *gal*
MID Canal	Surface water	0.83	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		20.77	3.98	52.66	211.51	

16 - 06/16/2023: Corn, silage

Field name: 16

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.97	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		4.97	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.72	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.72	0.00	0.00	0.00	

16 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.92	25.65	129.91	1,015.16	8,219,520.00 *gal*
MID Canal	Surface water	4.97	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		127.88	25.65	129.91	1,015.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.97	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		4.97	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.72	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.72	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.86	19.17	97.09	758.70	6,143,040.00 *gal*
MID Canal	Surface water	3.72	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		95.59	19.17	97.09	758.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.72	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.72	0.00	0.00	0.00	

17 - 04/23/2023: Corn, silage

Field name: 17

Crop: Corn, silage

Plant date: 04/23/2023

17 - 04/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	5.54	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		5.54	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	206.21	41.19	544.41	2,186.71	1,900,800.00 *gal*
MID Canal	Surface water	3.70	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		209.90	41.19	544.41	2,186.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.96	0.00	0.00	0.00	3,456,000.00 *gal*
Application event totals		2.96	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.51	0.00	0.00	0.00	4,104,000.00 *gal*
Application event totals		3.51	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.96	0.00	0.00	0.00	3,456,000.00 *gal*
Application event totals		2.96	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.33	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		3.33	0.00	0.00	0.00	

17 - 04/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.96	0.00	0.00	0.00	3,456,000.00 *gal*
Application event totals		2.96	0.00	0.00	0.00	

18 - 11/03/2022: WheatHay

Field name: 18

Crop: WheatHay Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/05/2022	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.11	12.99	94.50	605.48	862,920.00 *gal*
MID Canal	Surface water	2.53	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		108.64	12.99	94.50	605.48	

18 - 07/29/2023: Sudangrass, silage

Field name: 18

Crop: Sudangrass, silage Plant date: 07/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.80	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.80	0.00	0.00	0.00	

18 - 07/29/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.71	7.66	38.79	303.14	2,405,400.00 *gal*
MID Canal	Surface water	2.01	0.00	0.00	0.00	4,104,000.00 *gal*
Application event totals		38.71	7.66	38.79	303.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.17	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		3.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.77	7.26	36.75	287.19	2,278,800.00 *gal*
MID Canal	Surface water	1.90	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		36.67	7.26	36.75	287.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.85	0.00	0.00	0.00	5,832,000.00 *gal*
Application event totals		2.85	0.00	0.00	0.00	

2 - 04/13/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 04/13/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 04/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.60	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		4.60	0.00	0.00	0.00	

05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.87	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		2.87	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	215.06	42.95	567.78	2,280.59	637,200.00 *gal*
MID Canal	Surface water	3.45	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		218.51	42.95	567.78	2,280.59	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.94	11.05	55.95	437.21	637,200.00 *gal*
MID Canal	Surface water	3.45	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		56.39	11.05	55.95	437.21	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

20 N - 04/24/2023: Corn, silage

Field name: 20 N

Crop: Corn, silage

Plant date: 04/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.43	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.55	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		3.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.66	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		2.66	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.43	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	244.95	48.92	646.70	2,597.55	2,822,400.00 *gal*
MID Canal	Surface water	4.43	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		249.38	48.92	646.70	2,597.55	

20 N - 04/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.43	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.43	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.50	9.91	50.20	392.28	2,223,360.00 *gal*
MID Canal	Surface water	3.55	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		51.05	9.91	50.20	392.28	

21 - 04/23/2023: Corn, silage

Field name: 21

Crop: Corn, silage

Plant date: 04/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.10	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.14	0.00	0.00	0.00	2,592,000.00 *gal*
Application event totals		4.14	0.00	0.00	0.00	

21 - 04/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.10	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.10	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.10	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.76	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		2.76	0.00	0.00	0.00	

22 - 04/20/2023: Corn, silage

Field name: 22

Crop: Corn, silage

Plant date: 04/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

22 - 04/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	1.62	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		1.62	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.16	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		2.16	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.60	0.00	0.00	0.00	17,280,000.00 *gal*
Application event totals		3.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.60	0.00	0.00	0.00	17,280,000.00 *gal*
Application event totals		3.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.36	23.04	304.57	1,223.35	4,367,520.00 *gal*
MID Canal	Surface water	1.62	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		116.98	23.04	304.57	1,223.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.79	0.00	0.00	0.00	13,392,000.00 *gal*
Application event totals		2.79	0.00	0.00	0.00	

22 - 04/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.79	0.00	0.00	0.00	13,392,000.00 *gal*
Application event totals		2.79	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.43	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.73	13.09	66.30	518.08	9,648,000.00 *gal*
MID Canal	Surface water	2.16	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		64.89	13.09	66.30	518.08	

24 E - 11/10/2022: Wheat, silage, soft dough

Field name: 24 E

Crop: Wheat, silage, soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.52	15.25	110.89	710.55	1,674,000.00 *gal*
MID Canal	Surface water	2.43	0.00	0.00	0.00	8,208,000.00 *gal*
Application event totals		126.95	15.25	110.89	710.55	

24 E - 11/10/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.98	6.34	59.85	740.71	1,516,800.00 *gal*
MID Canal	Surface water	2.04	0.00	0.00	0.00	6,912,000.00 *gal*
Application event totals		90.02	6.34	59.85	740.71	

24 E - 06/08/2023: Corn, silage

Field name: 24 E

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.83	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		3.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.96	0.00	0.00	0.00	13,392,000.00 *gal*
Application event totals		3.96	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.68	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		2.68	0.00	0.00	0.00	

24 E - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	119.63	24.97	126.44	988.05	12,960,000.00 *gal*
MID Canal	Surface water	3.83	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		123.47	24.97	126.44	988.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.07	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.07	0.00	0.00	0.00	

25 - 06/11/2023: Corn, silage

Field name: 25

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.05	0.00	0.00	0.00	15,552,000.00 *gal*
Application event totals		4.05	0.00	0.00	0.00	

25 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	120.00	0.00	0.00	0.00	
Application event totals		120.00	0.00	0.00	0.00	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.37	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		3.37	0.00	0.00	0.00	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.71	0.00	0.00	0.00	14,256,000.00 *gal*
Application event totals		3.71	0.00	0.00	0.00	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.04	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.04	0.00	0.00	0.00	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.04	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.04	0.00	0.00	0.00	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.70	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		2.70	0.00	0.00	0.00	

25 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.70	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		2.70	0.00	0.00	0.00	

26 - 11/11/2022: Wheat, silage, soft dough

Field name: 26

Crop: Wheat, silage, soft dough

Plant date: 11/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	208.06	80.56	31.73	0.00	400.00 *ton*
Application event totals		208.06	80.56	31.73	0.00	

03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.19	0.00	0.00	0.00	12,960,000.00 *gal*
Application event totals		4.19	0.00	0.00	0.00	

26 - 06/02/2023: Corn, silage

Field name: 26

Crop: Corn, silage

Plant date: 06/02/2023

26 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	5.03	0.00	0.00	0.00	15,552,000.00 *gal*
Application event totals		5.03	0.00	0.00	0.00	

06/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	120.00	0.00	0.00	0.00	
Application event totals		120.00	0.00	0.00	0.00	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.78	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.78	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.78	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.78	0.00	0.00	0.00	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

26 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.36	0.00	0.00	0.00	

3 - 04/15/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 04/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	144.54	28.87	381.61	1,532.81	1,570,320.00 *gal*
MID Canal	Surface water	3.76	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		148.31	28.87	381.61	1,532.81	

05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.70	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.70	0.00	0.00	0.00	

3 - 04/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	151.30	30.22	399.46	1,604.49	1,643,760.00 *gal*
MID Canal	Surface water	2.82	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		154.13	30.22	399.46	1,604.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.70	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.70	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.70	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.70	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.70	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.70	0.00	0.00	0.00	

4 - 04/14/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 04/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.27	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.27	0.00	0.00	0.00	

4 - 04/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.27	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.27	0.00	0.00	0.00	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.68	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.68	0.00	0.00	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	177.04	35.36	467.40	1,877.37	4,429,440.00 *gal*
MID Canal	Surface water	3.27	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		180.30	35.36	467.40	1,877.37	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.27	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.27	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.25	13.83	70.02	547.12	6,733,440.00 *gal*
MID Canal	Surface water	3.27	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		69.51	13.83	70.02	547.12	

4 - 04/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.39	10.10	51.14	399.63	4,918,320.00 *gal*
MID Canal	Surface water	3.68	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		52.06	10.10	51.14	399.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.68	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.68	0.00	0.00	0.00	

4 - 08/14/2023: Sudangrass, silage

Field name: 4

Crop: Sudangrass, silage Plant date: 08/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.68	0.00	0.00	0.00	11,664,000.00 *gal*
Application event totals		3.68	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.40	8.64	43.76	341.95	4,208,400.00 *gal*
MID Canal	Surface water	2.04	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		43.45	8.64	43.76	341.95	

4 - 08/14/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.27	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		3.27	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.68	10.37	52.51	410.34	5,050,080.00 *gal*
MID Canal	Surface water	2.45	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		52.13	10.37	52.51	410.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.04	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		2.04	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.45	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		2.45	0.00	0.00	0.00	

5 - 04/17/2002: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 04/17/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	5.17	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		5.17	0.00	0.00	0.00	

5 - 04/17/2002: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.31	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		4.31	0.00	0.00	0.00	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	169.30	33.81	446.98	1,795.37	2,006,520.00 *gal*
MID Canal	Surface water	3.31	0.00	0.00	0.00	4,968,000.00 *gal*
Application event totals		172.61	33.81	446.98	1,795.37	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.80	16.14	213.32	856.83	957,600.00 *gal*
MID Canal	Surface water	3.45	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		84.25	16.14	213.32	856.83	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.53	6.79	34.38	268.63	1,566,000.00 *gal*
MID Canal	Surface water	2.59	0.00	0.00	0.00	3,888,000.00 *gal*
Application event totals		35.11	6.79	34.38	268.63	

5 - 04/17/2002: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

6 - 04/18/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 04/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.36	0.00	0.00	0.00	10,368,000.00 *gal*
Application event totals		4.36	0.00	0.00	0.00	

05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.81	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.58	22.29	294.58	1,183.22	2,093,760.00 *gal*
MID Canal	Surface water	2.18	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		113.76	22.29	294.58	1,183.22	

06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.27	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.27	0.00	0.00	0.00	

6 - 04/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	114.79	22.93	303.05	1,217.26	2,154,000.00 *gal*
MID Canal	Surface water	2.45	0.00	0.00	0.00	5,832,000.00 *gal*
Application event totals		117.24	22.93	303.05	1,217.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.09	0.00	0.00	0.00	7,344,000.00 *gal*
Application event totals		3.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.90	0.00	0.00	0.00	6,912,000.00 *gal*
Application event totals		2.90	0.00	0.00	0.00	

6 - 08/14/2023: Sudangrass, silage

Field name: 6

Crop: Sudangrass, silage

Plant date: 08/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.81	0.00	0.00	0.00	9,072,000.00 *gal*
Application event totals		3.81	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.00	0.00	0.00	0.00	7,128,000.00 *gal*
Application event totals		3.00	0.00	0.00	0.00	

6 - 08/14/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.00	17.74	89.84	702.03	6,480,000.00 *gal*
MID Canal	Surface water	2.72	0.00	0.00	0.00	6,480,000.00 *gal*
Application event totals		87.73	17.74	89.84	702.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.27	0.00	0.00	0.00	7,776,000.00 *gal*
Application event totals		3.27	0.00	0.00	0.00	

7 - 04/20/2023: Corn, silage

Field name: 7

Crop: Corn, silage Plant date: 04/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.78	0.00	0.00	0.00	2,592,000.00 *gal*
Application event totals		4.78	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.98	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		3.98	0.00	0.00	0.00	

7 - 04/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	122.31	24.43	322.90	1,296.99	523,440.00 *gal*
MID Canal	Surface water	2.39	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		124.69	24.43	322.90	1,296.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.58	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.58	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.88	22.34	295.37	1,186.38	478,800.00 *gal*
MID Canal	Surface water	2.39	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		114.26	22.34	295.37	1,186.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.58	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.58	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.58	0.00	0.00	0.00	1,944,000.00 *gal*
Application event totals		3.58	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.18	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		3.18	0.00	0.00	0.00	

8 - 11/02/2022: Wheat, silage, soft dough

Field name: 8

Crop: Wheat, silage, soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	182.74	36.50	482.46	1,937.86	541,440.00 *gal*
MID Canal	Surface water	2.30	0.00	0.00	0.00	864,000.00 *gal*
Application event totals		185.04	36.50	482.46	1,937.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

8 - 07/29/2023: Sudangrass, silage

Field name: 8

Crop: Sudangrass, silage

Plant date: 07/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.60	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		4.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.45	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		3.45	0.00	0.00	0.00	

8 - 07/29/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	4.60	0.00	0.00	0.00	1,728,000.00 *gal*
Application event totals		4.60	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.46	23.47	118.86	928.77	1,353,600.00 *gal*
Application event totals		112.46	23.47	118.86	928.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	2.87	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		2.87	0.00	0.00	0.00	

Pistachios - 01/01/2022: Pistachios

Field name: Pistachios

Crop: Pistachios

Plant date: 01/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.34	0.00	0.00	0.00	64,800,000.00 *gal*
Application event totals		3.34	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.46	0.00	0.00	0.00	66,960,000.00 *gal*
Application event totals		3.46	0.00	0.00	0.00	

Pistachios - 01/01/2022: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
10-0-8	Solid commercial fertilizer	115.00	0.00	92.00	0.00	
Application event totals		115.00	0.00	92.00	0.00	

06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.34	0.00	0.00	0.00	64,800,000.00 *gal*
Application event totals		3.34	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.46	0.00	0.00	0.00	66,960,000.00 *gal*
Application event totals		3.46	0.00	0.00	0.00	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.46	0.00	0.00	0.00	66,960,000.00 *gal*
Application event totals		3.46	0.00	0.00	0.00	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.34	0.00	0.00	0.00	64,800,000.00 *gal*
Application event totals		3.34	0.00	0.00	0.00	

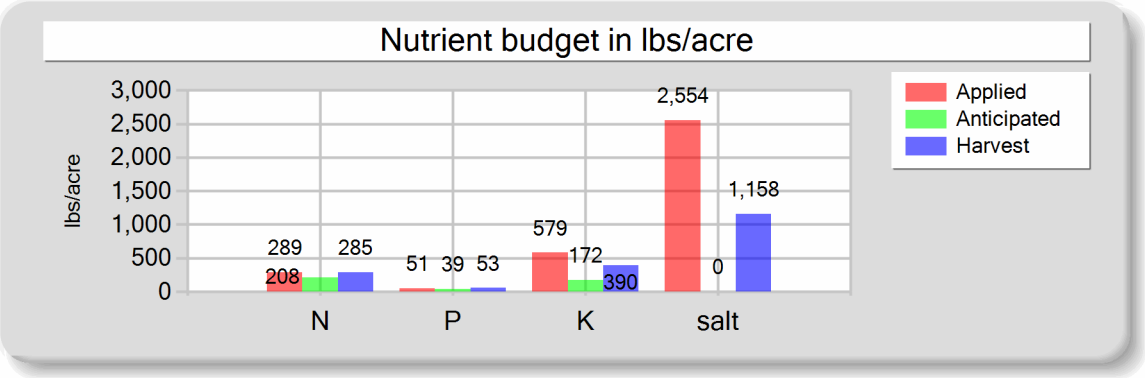
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Canal	Surface water	3.46	0.00	0.00	0.00	66,960,000.00 *gal*
Application event totals		3.46	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 04/12/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 04/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	253.36	51.11	579.41	2,554.17
Fresh water	21.83	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	289.20	51.11	579.41	2,554.17
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	285.30	53.41	390.01	1,158.07
Nutrient balance	3.89	-2.30	189.41	1,396.10
Applied to removed ratio	1.01	0.96	1.49	2.21

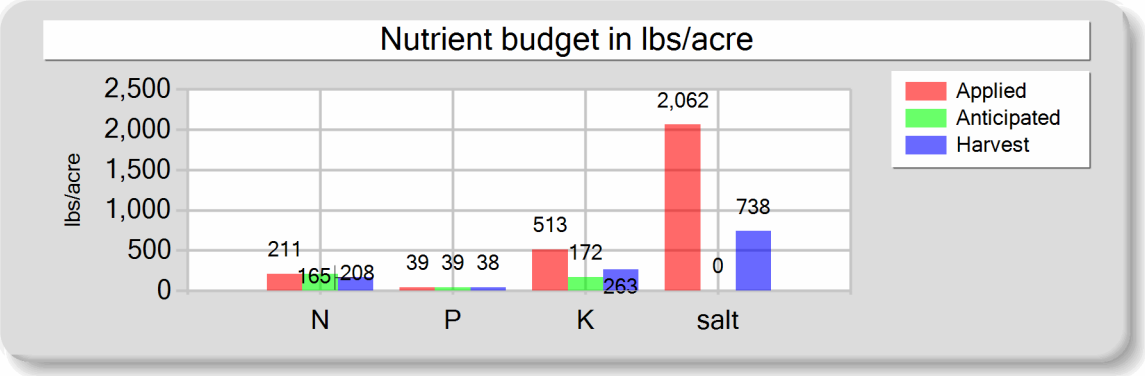
Fresh water applied	
53,784,000.00	*gallons*
1,980.68	*acre-inches*
33.57	*inches/acre*

Process wastewater applied	
8,283,600.00	*gallons*
305.06	*acre-inches*
5.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 04/27/2023: Corn, silage

Field name: 12 Crop: Corn, silage Plant date: 04/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.41	38.83	513.26	2,061.60
Fresh water	9.78	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.19	38.83	513.26	2,061.60
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	165.44	38.41	263.43	737.60
Nutrient balance	45.75	0.42	249.84	1,324.00
Applied to removed ratio	1.28	1.01	1.95	2.80

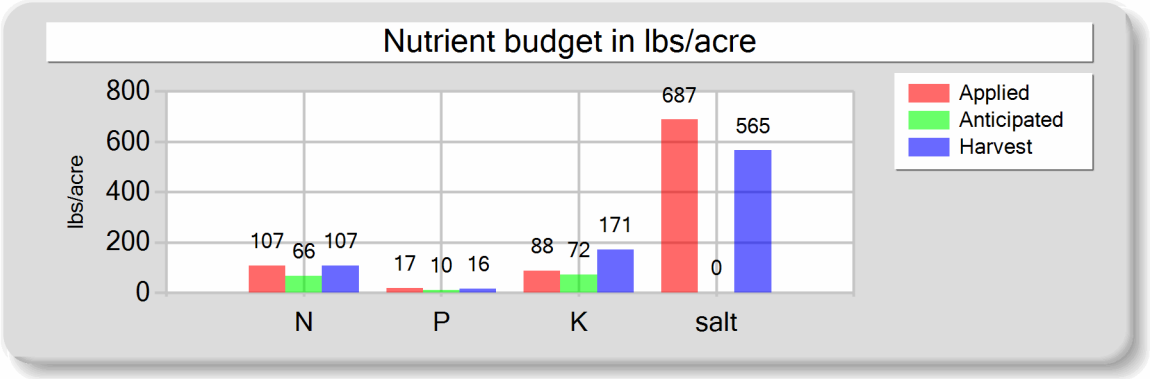
Fresh water applied	
37,584,000.00	*gallons*
1,384.09	*acre-inches*
15.04	*inches/acre*

Process wastewater applied	
5,888,160.00	*gallons*
216.84	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 08/15/2023: Sudangrass, silage

Field name: 12 Crop: Sudangrass, silage Plant date: 08/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	83.23	17.37	87.97	687.38
Fresh water	16.98	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	107.21	17.37	87.97	687.38
Anticipated crop nutrient removal	66.00	10.20	72.00	0.00
Actual crop nutrient removal	106.61	16.32	171.15	565.02
Nutrient balance	0.60	1.05	-83.19	122.36
Applied to removed ratio	1.01	1.06	0.51	1.22

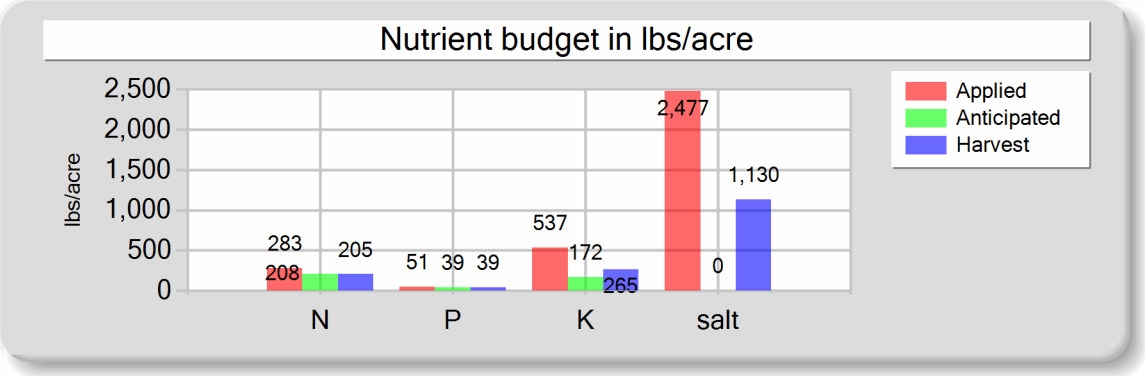
Fresh water applied	
32,616,000.00	*gallons*
1,201.14	*acre-inches*
26.11	*inches/acre*

Process wastewater applied	
5,120,280.00	*gallons*
188.56	*acre-inches*
4.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 04/25/2023: Corn, silage

Field name: 13 Crop: Corn, silage Plant date: 04/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	251.17	50.88	536.99	2,476.62
Fresh water	24.36	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.53	50.88	536.99	2,476.62
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	205.11	39.28	265.48	1,130.29
Nutrient balance	77.42	11.60	271.51	1,346.34
Applied to removed ratio	1.38	1.30	2.02	2.19

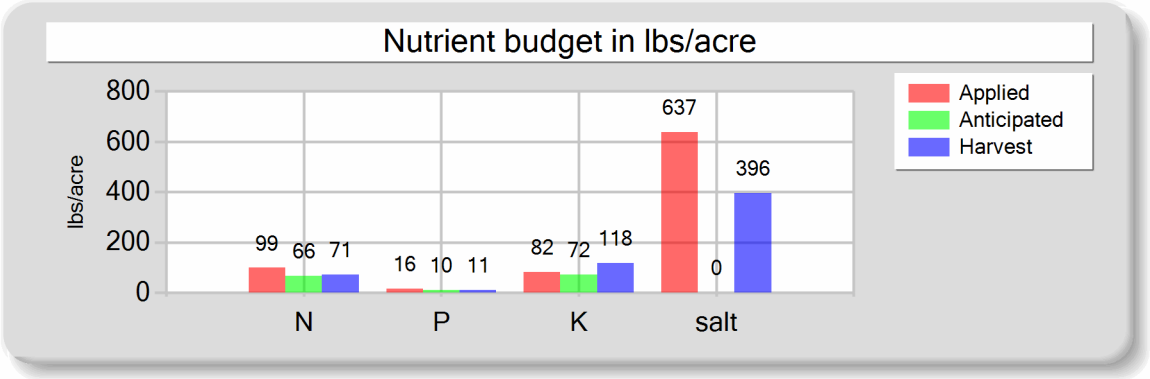
Fresh water applied	
80,352,000.00	*gallons*
2,959.09	*acre-inches*
37.46	*inches/acre*

Process wastewater applied	
12,877,200.00	*gallons*
474.22	*acre-inches*
6.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

13 - 08/16/2023: Sudangrass, silage

Field name: 13 Crop: Sudangrass, silage Plant date: 08/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	77.18	16.11	81.57	637.38
Fresh water	14.54	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	98.71	16.11	81.57	637.38
Anticipated crop nutrient removal	66.00	10.20	72.00	0.00
Actual crop nutrient removal	71.47	11.41	118.07	395.58
Nutrient balance	27.24	4.70	-36.50	241.80
Applied to removed ratio	1.38	1.41	0.69	1.61

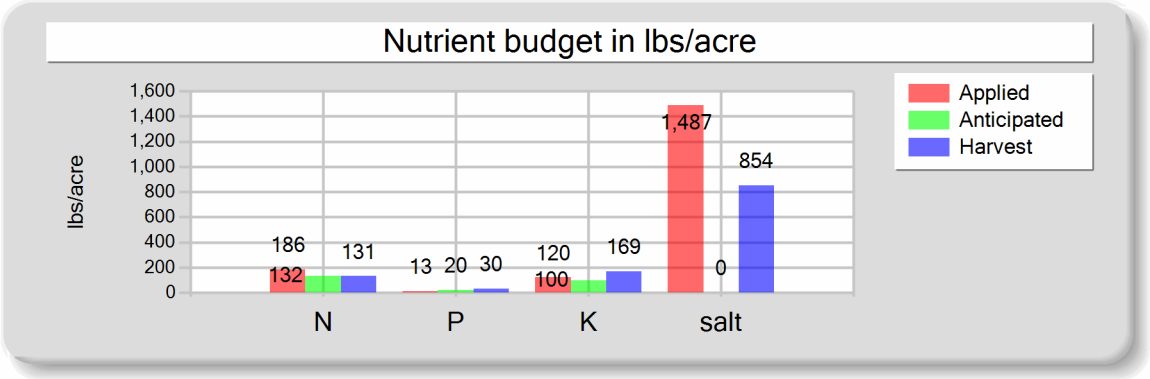
Fresh water applied	
47,952,000.00	*gallons*
1,765.91	*acre-inches*
22.35	*inches/acre*

Process wastewater applied	
8,154,000.00	*gallons*
300.28	*acre-inches*
3.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 11/06/2022: Wheat Hay

Field name: 14 Crop: Wheat Hay Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	176.68	12.74	120.19	1,487.49
Fresh water	2.10	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.78	12.74	120.19	1,487.49
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	130.89	29.63	168.51	853.54
Nutrient balance	54.88	-16.89	-48.32	633.95
Applied to removed ratio	1.42	0.43	0.71	1.74

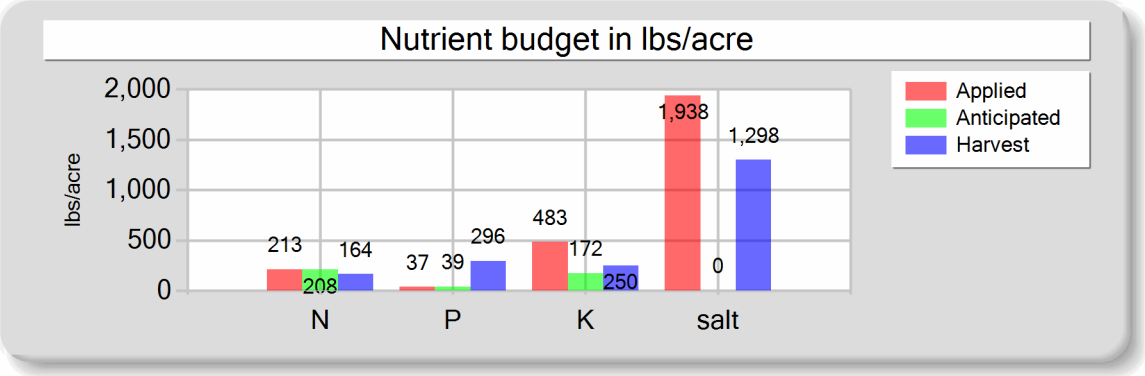
Fresh water applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
3.22	*inches/acre*

Process wastewater applied	
2,782,800.00	*gallons*
102.48	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

14 - 06/09/2023: Corn, silage

Field name: 14 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.77	36.50	482.53	1,938.13
Fresh water	23.07	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.84	36.50	482.53	1,938.13
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	163.59	296.35	250.12	1,298.02
Nutrient balance	49.25	-259.85	232.40	640.12
Applied to removed ratio	1.30	0.12	1.93	1.49

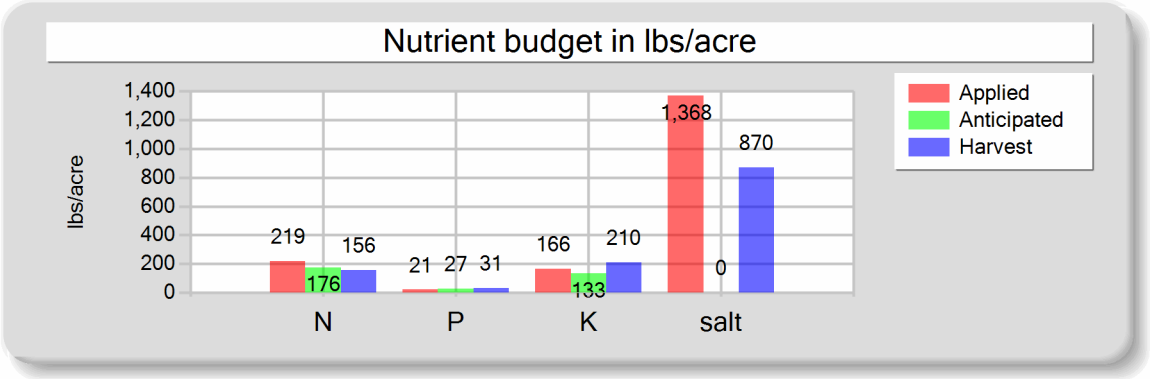
Fresh water applied	
71,280,000.00	*gallons*
2,625.00	*acre-inches*
35.47	*inches/acre*

Process wastewater applied	
4,452,480.00	*gallons*
163.97	*acre-inches*
2.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

15 - 11/09/2022: Wheat, silage, soft dough

Field name: 15 Crop: Wheat, silage, soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.08	21.21	165.95	1,368.36
Fresh water	7.66	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.74	21.21	165.95	1,368.36
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	155.72	31.03	209.52	870.42
Nutrient balance	63.02	-9.82	-43.57	497.94
Applied to removed ratio	1.40	0.68	0.79	1.57

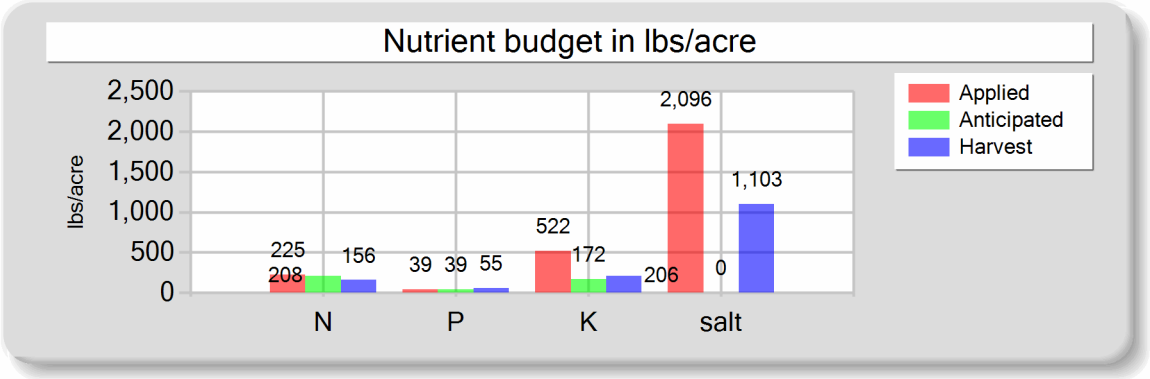
Fresh water applied	
24,624,000.00	*gallons*
906.82	*acre-inches*
11.78	*inches/acre*

Process wastewater applied	
2,879,280.00	*gallons*
106.03	*acre-inches*
1.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

15 - 06/19/2023: Corn, silage

Field name: 15 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.69	39.48	521.92	2,096.35
Fresh water	20.67	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.36	39.48	521.92	2,096.35
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	155.96	54.66	206.37	1,102.54
Nutrient balance	69.40	-15.17	315.54	993.82
Applied to removed ratio	1.45	0.72	2.53	1.90

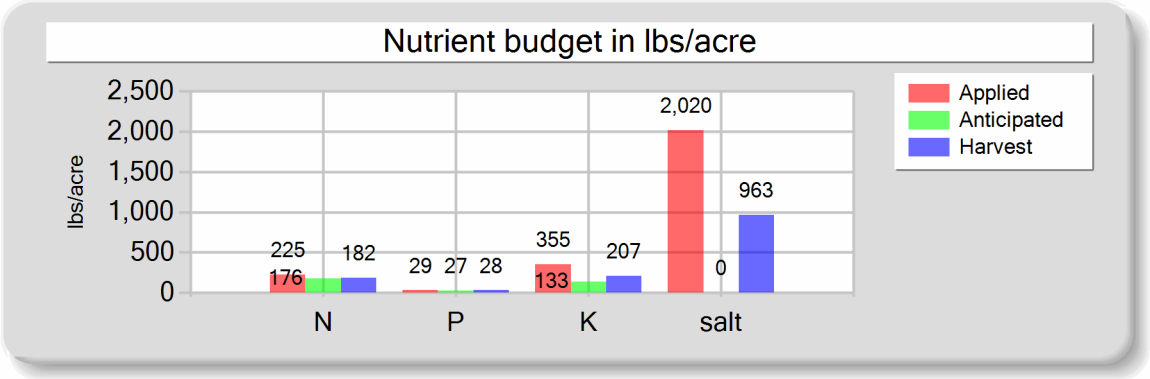
Fresh water applied	
66,464,000.00	*gallons*
2,447.64	*acre-inches*
31.79	*inches/acre*

Process wastewater applied	
5,011,200.00	*gallons*
184.55	*acre-inches*
2.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

16 - 11/08/2022: Wheat, silage, soft dough

Field name: 16 Crop: Wheat, silage, soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.02	29.00	354.78	2,019.81
Fresh water	5.79	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.82	29.00	354.78	2,019.81
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	181.81	28.34	206.64	962.84
Nutrient balance	43.01	0.66	148.13	1,056.97
Applied to removed ratio	1.24	1.02	1.72	2.10

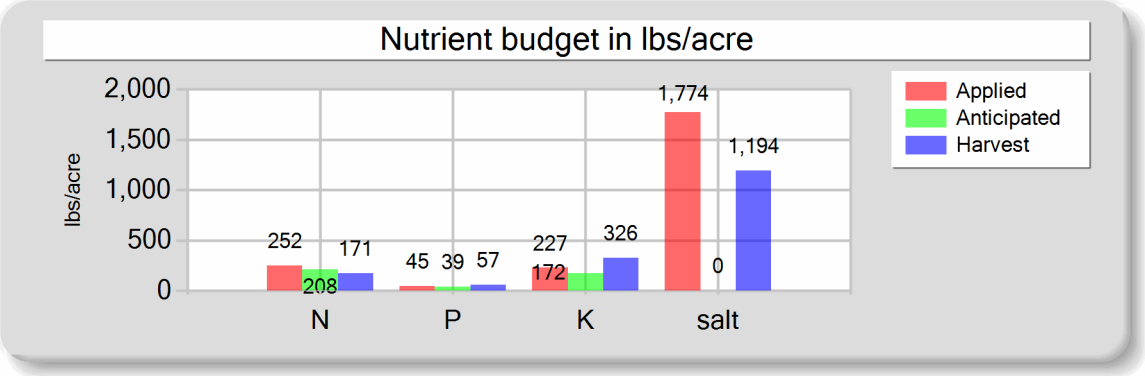
Fresh water applied	
12,096,000.00	*gallons*
445.45	*acre-inches*
8.91	*inches/acre*

Process wastewater applied	
2,881,440.00	*gallons*
106.11	*acre-inches*
2.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

16 - 06/16/2023: Corn, silage

Field name: 16 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.78	44.83	227.01	1,773.86
Fresh water	29.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.58	44.83	227.01	1,773.86
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	171.13	57.04	325.60	1,194.36
Nutrient balance	80.45	-12.22	-98.59	579.51
Applied to removed ratio	1.47	0.79	0.70	1.49

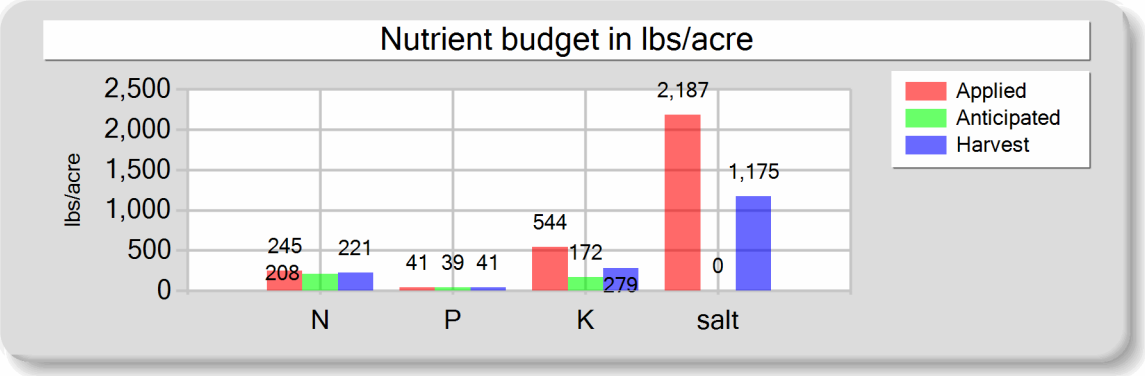
Fresh water applied	
62,208,000.00	*gallons*
2,290.91	*acre-inches*
45.82	*inches/acre*

Process wastewater applied	
14,362,560.00	*gallons*
528.92	*acre-inches*
10.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

17 - 04/23/2023: Corn, silage

Field name: 17 Crop: Corn, silage Plant date: 04/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.21	41.19	544.41	2,186.71
Fresh water	24.94	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	245.15	41.19	544.41	2,186.71
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	221.49	40.80	278.61	1,174.88
Nutrient balance	23.66	0.39	265.81	1,011.84
Applied to removed ratio	1.11	1.01	1.95	1.86

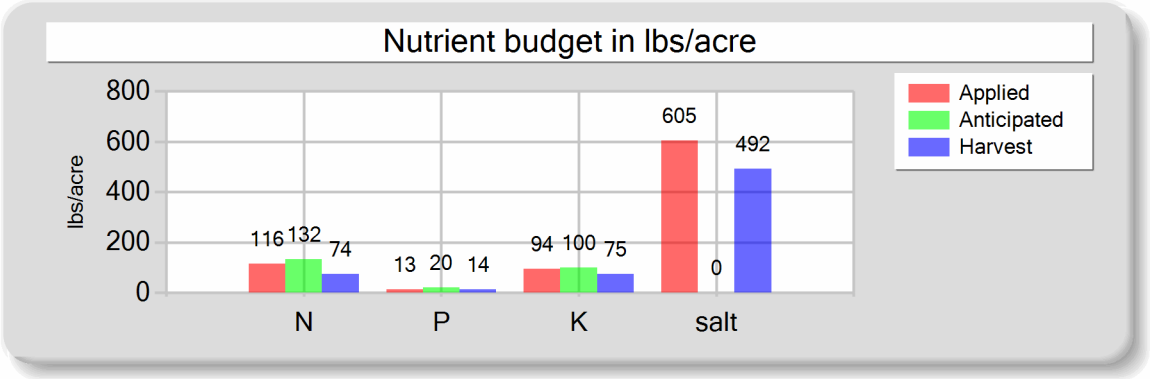
Fresh water applied	
29,160,000.00	*gallons*
1,073.86	*acre-inches*
38.35	*inches/acre*

Process wastewater applied	
1,900,800.00	*gallons*
70.00	*acre-inches*
2.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

18 - 11/03/2022: WheatHay

Field name: 18 Crop: WheatHay Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	106.11	12.99	94.50	605.48
Fresh water	2.53	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	115.64	12.99	94.50	605.48
Anticipated crop nutrient removal	132.00	20.40	99.60	0.00
Actual crop nutrient removal	73.54	14.14	74.80	492.17
Nutrient balance	42.10	-1.15	19.70	113.31
Applied to removed ratio	1.57	0.92	1.26	1.23

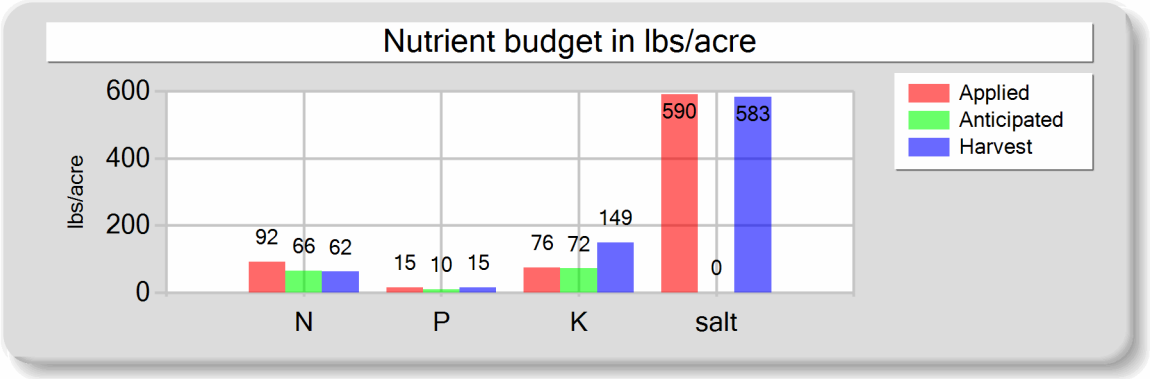
Fresh water applied	
5,184,000.00	*gallons*
190.91	*acre-inches*
3.90	*inches/acre*

Process wastewater applied	
862,920.00	*gallons*
31.78	*acre-inches*
0.65	*inches/acre*

Total harvests for the crop	
1	*harvests*

18 - 07/29/2023: Sudangrass, silage

Field name: 18 Crop: Sudangrass, silage Plant date: 07/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	71.48	14.92	75.55	590.33
Fresh water	13.72	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	92.20	14.92	75.55	590.33
Anticipated crop nutrient removal	66.00	10.20	72.00	0.00
Actual crop nutrient removal	62.35	15.36	148.96	583.29
Nutrient balance	29.85	-0.44	-73.41	7.04
Applied to removed ratio	1.48	0.97	0.51	1.01

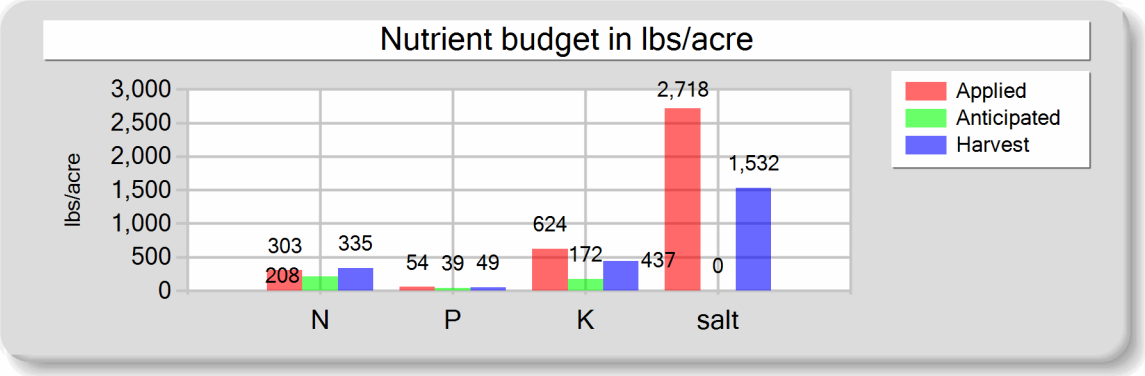
Fresh water applied	
28,080,000.00	*gallons*
1,034.09	*acre-inches*
21.10	*inches/acre*

Process wastewater applied	
4,684,200.00	*gallons*
172.50	*acre-inches*
3.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 04/13/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 04/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	268.00	54.00	623.74	2,717.80
Fresh water	21.27	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	303.27	54.00	623.74	2,717.80
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	334.56	48.84	437.33	1,531.83
Nutrient balance	-31.29	5.17	186.40	1,185.97
Applied to removed ratio	0.91	1.11	1.43	1.77

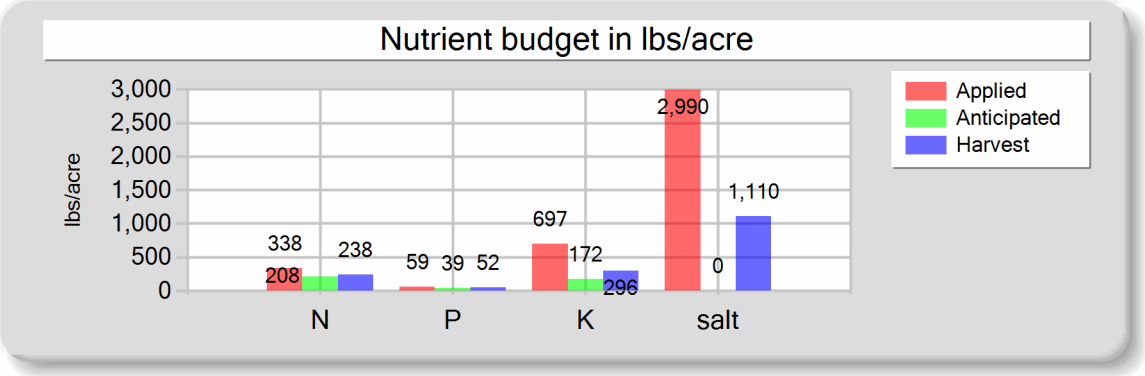
Fresh water applied	
7,992,000.00	*gallons*
294.32	*acre-inches*
32.70	*inches/acre*

Process wastewater applied	
1,274,400.00	*gallons*
46.93	*acre-inches*
5.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

20 N - 04/24/2023: Corn, silage

Field name: 20 N Crop: Corn, silage Plant date: 04/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.45	58.84	696.90	2,989.83
Fresh water	31.93	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	338.37	58.84	696.90	2,989.83
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	237.66	51.76	295.54	1,110.47
Nutrient balance	100.72	7.07	401.36	1,879.36
Applied to removed ratio	1.42	1.14	2.36	2.69

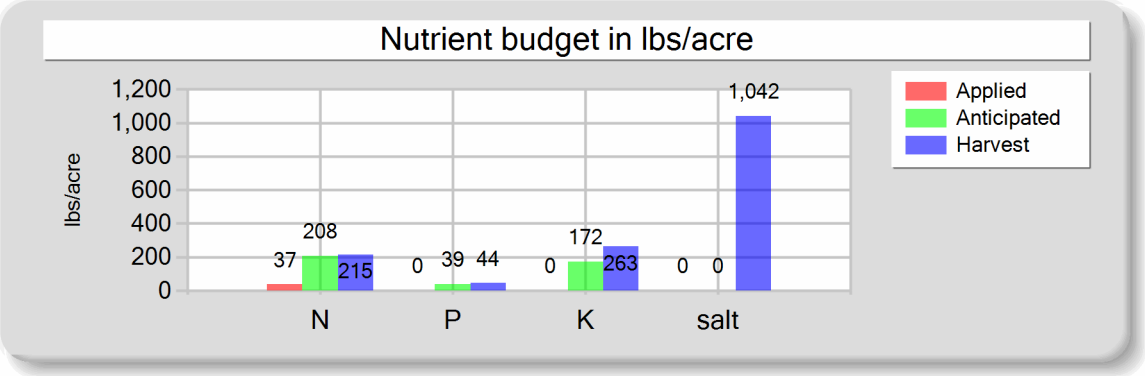
Fresh water applied	
46,656,000.00	*gallons*
1,718.18	*acre-inches*
49.09	*inches/acre*

Process wastewater applied	
5,045,760.00	*gallons*
185.82	*acre-inches*
5.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

21 - 04/23/2023: Corn, silage

Field name: 21 Crop: Corn, silage Plant date: 04/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.76	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	36.76	0.00	0.00	0.00
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	215.33	44.43	263.39	1,042.21
Nutrient balance	-178.57	-44.43	-263.39	-1,042.21
Applied to removed ratio	0.17	0.00	0.00	0.00

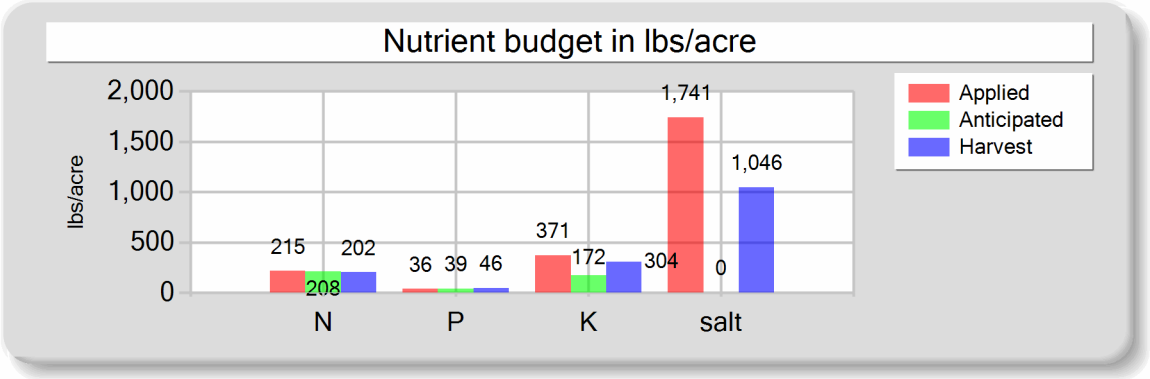
Fresh water applied	
14,256,000.00	*gallons*
525.00	*acre-inches*
35.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

22 - 04/20/2023: Corn, silage

Field name: 22 Crop: Corn, silage Plant date: 04/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	178.09	36.13	370.87	1,741.43
Fresh water	22.76	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	214.85	36.13	370.87	1,741.43
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	201.69	45.89	304.13	1,046.21
Nutrient balance	13.17	-9.75	66.74	695.22
Applied to removed ratio	1.07	0.79	1.22	1.66

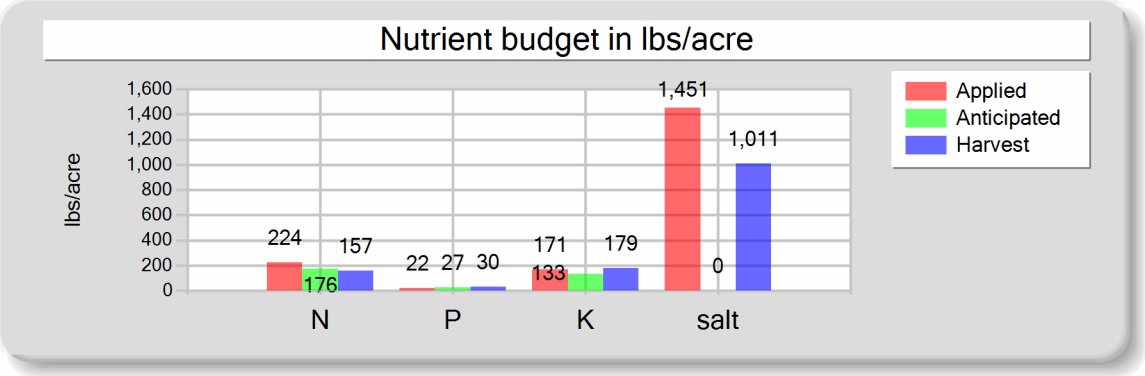
Fresh water applied	
109,296,000.00	*gallons*
4,025.00	*acre-inches*
35.00	*inches/acre*

Process wastewater applied	
14,015,520.00	*gallons*
516.14	*acre-inches*
4.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

24 E - 11/10/2022: Wheat, silage, soft dough

Field name: 24 E | Crop: Wheat, silage, soft dough | Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.50	21.59	170.74	1,451.26
Fresh water	4.47	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.97	21.59	170.74	1,451.26
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	157.08	29.96	179.48	1,010.80
Nutrient balance	66.89	-8.37	-8.74	440.46
Applied to removed ratio	1.43	0.72	0.95	1.44

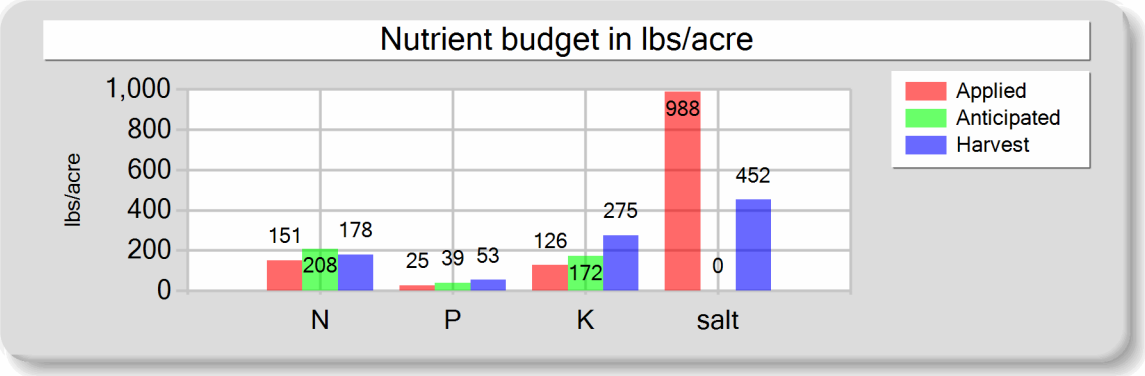
Fresh water applied	
15,120,000.00	*gallons*
556.82	*acre-inches*
6.87	*inches/acre*

Process wastewater applied	
3,190,800.00	*gallons*
117.51	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

24 E - 06/08/2023: Corn, silage

Field name: 24 E Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	119.63	24.97	126.44	988.05
Fresh water	24.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.90	24.97	126.44	988.05
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	178.18	53.33	275.43	451.75
Nutrient balance	-27.28	-28.36	-148.99	536.30
Applied to removed ratio	0.85	0.47	0.46	2.19

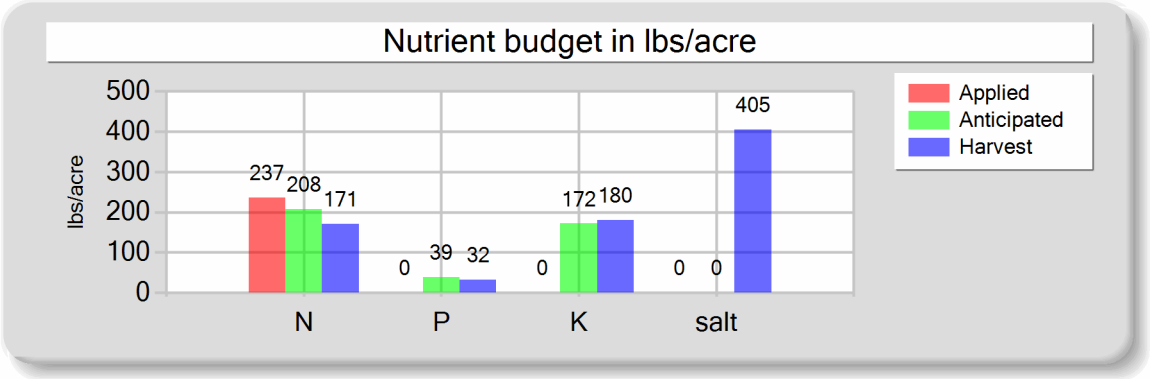
Fresh water applied	
82,080,000.00	*gallons*
3,022.73	*acre-inches*
37.32	*inches/acre*

Process wastewater applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
5.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

25 - 06/11/2023: Corn, silage

Field name: 25 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.60	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	236.60	0.00	0.00	0.00
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	170.64	32.39	179.73	404.80
Nutrient balance	65.96	-32.39	-179.73	-404.80
Applied to removed ratio	1.39	0.00	0.00	0.00

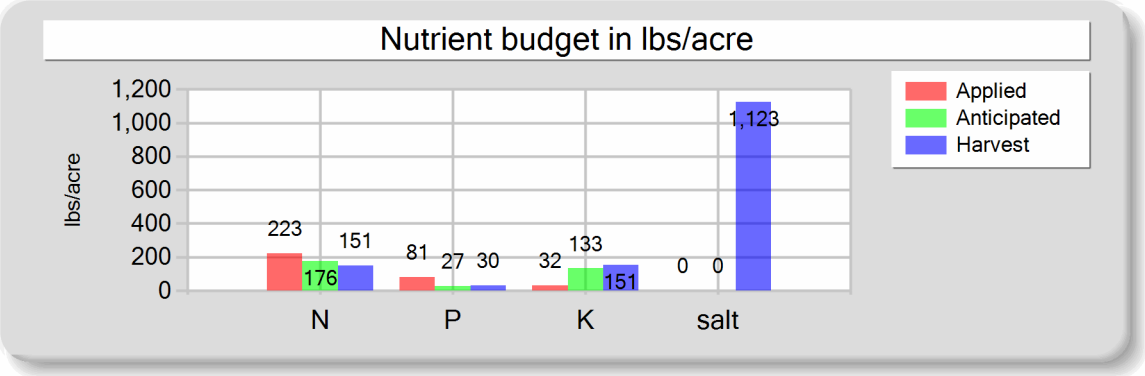
Fresh water applied	
86,832,000.00	*gallons*
3,197.73	*acre-inches*
34.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

26 - 11/11/2022: Wheat, silage, soft dough

Field name: 26 Crop: Wheat, silage, soft dough Plant date: 11/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	208.06	80.56	31.73	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	7.55	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.61	80.56	31.73	0.00
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	151.17	29.89	151.38	1,122.65
Nutrient balance	71.44	50.67	-119.65	-1,122.65
Applied to removed ratio	1.47	2.70	0.21	0.00

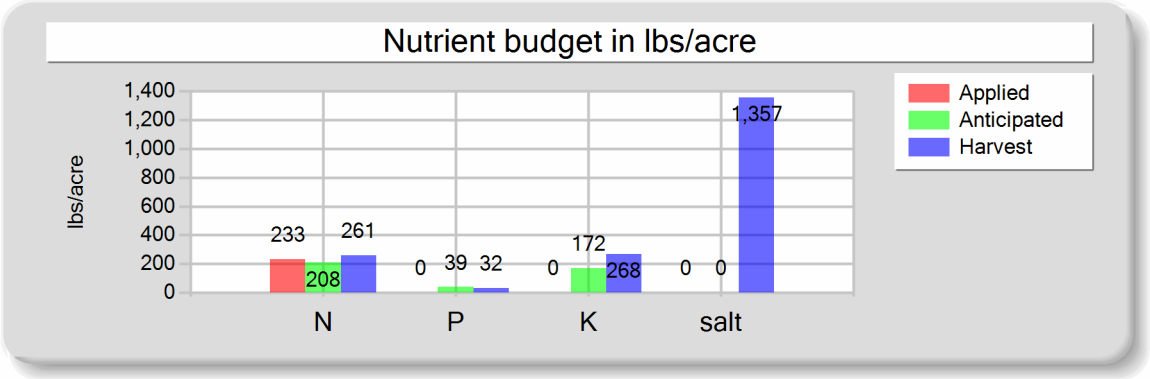
Fresh water applied	
23,328,000.00	*gallons*
859.09	*acre-inches*
11.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

26 - 06/02/2023: Corn, silage

Field name: 26 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	26.01	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.01	0.00	0.00	0.00
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	261.17	32.23	268.39	1,357.49
Nutrient balance	-28.16	-32.23	-268.39	-1,357.49
Applied to removed ratio	0.89	0.00	0.00	0.00

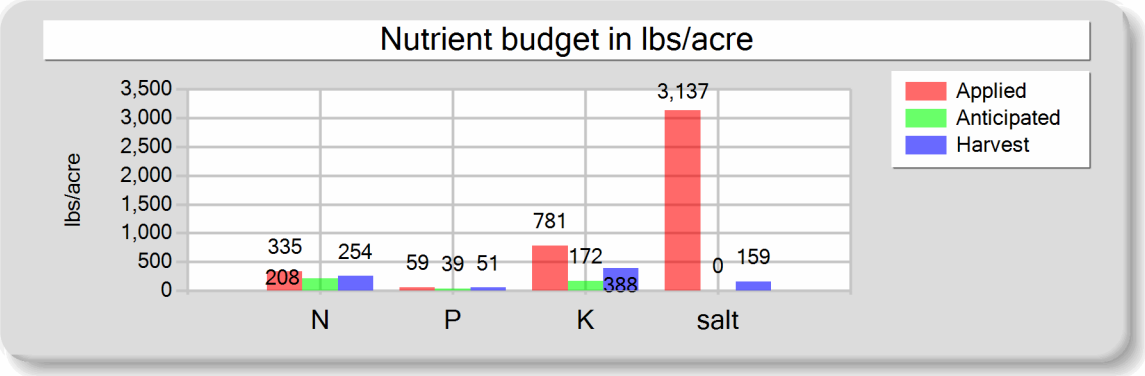
Fresh water applied
80,352,000.00 *gallons*
2,959.09 *acre-inches*
39.99 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3 - 04/15/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 04/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.85	59.09	781.07	3,137.30
Fresh water	25.40	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	335.25	59.09	781.07	3,137.30
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	253.85	50.77	387.94	159.29
Nutrient balance	81.39	8.32	393.13	2,978.01
Applied to removed ratio	1.32	1.16	2.01	19.70

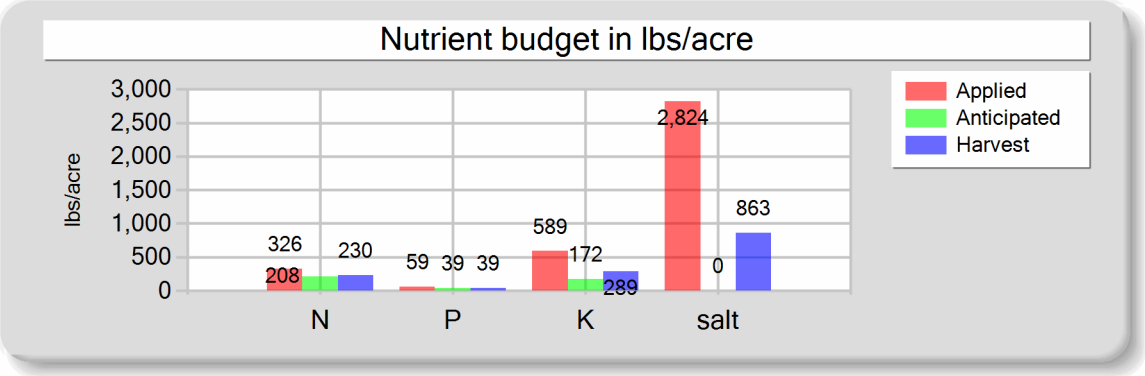
Fresh water applied	
34,992,000.00	*gallons*
1,288.64	*acre-inches*
39.05	*inches/acre*

Process wastewater applied	
3,214,080.00	*gallons*
118.36	*acre-inches*
3.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 04/14/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 04/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.67	59.28	588.55	2,824.12
Fresh water	27.36	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	326.03	59.28	588.55	2,824.12
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	230.41	38.97	289.15	863.37
Nutrient balance	95.62	20.32	299.41	1,960.75
Applied to removed ratio	1.42	1.52	2.04	3.27

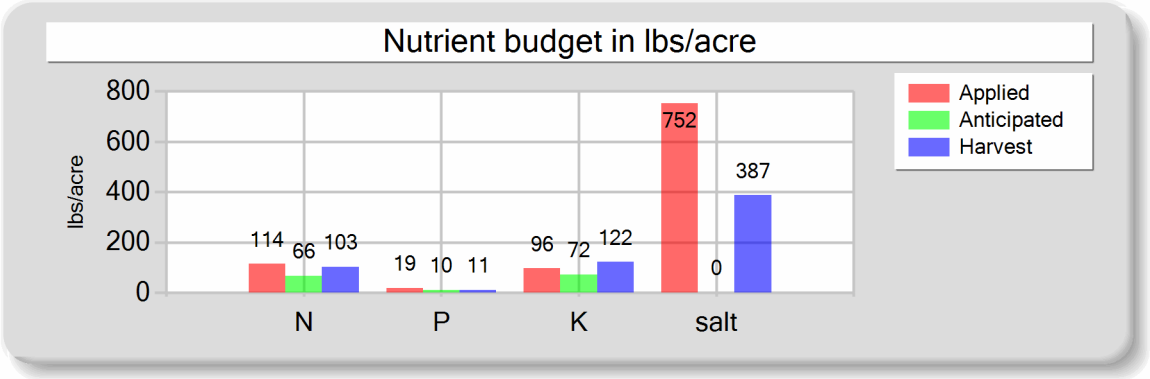
Fresh water applied	
86,832,000.00	*gallons*
3,197.73	*acre-inches*
42.08	*inches/acre*

Process wastewater applied	
16,081,200.00	*gallons*
592.22	*acre-inches*
7.79	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 08/14/2023: Sudangrass, silage

Field name: 4 Crop: Sudangrass, silage Plant date: 08/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	91.09	19.01	96.27	752.29
Fresh water	15.93	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	114.02	19.01	96.27	752.29
Anticipated crop nutrient removal	66.00	10.20	72.00	0.00
Actual crop nutrient removal	103.11	11.20	121.78	386.70
Nutrient balance	10.90	7.81	-25.50	365.59
Applied to removed ratio	1.11	1.70	0.79	1.95

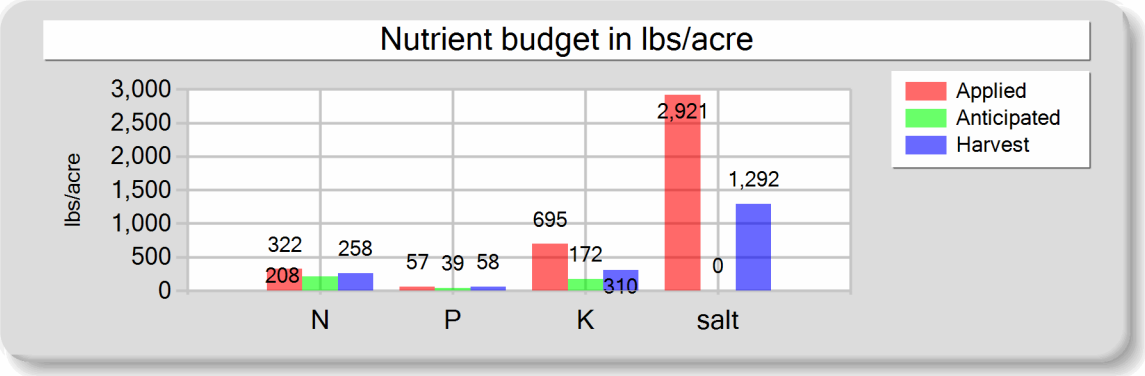
Fresh water applied	
50,544,000.00	*gallons*
1,861.36	*acre-inches*
24.49	*inches/acre*

Process wastewater applied	
9,258,480.00	*gallons*
340.96	*acre-inches*
4.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 04/17/2002: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 04/17/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	282.63	56.74	694.68	2,920.83
Fresh water	25.72	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	322.35	56.74	694.68	2,920.83
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	257.51	58.24	310.14	1,292.22
Nutrient balance	64.84	-1.50	384.54	1,628.61
Applied to removed ratio	1.25	0.97	2.24	2.26

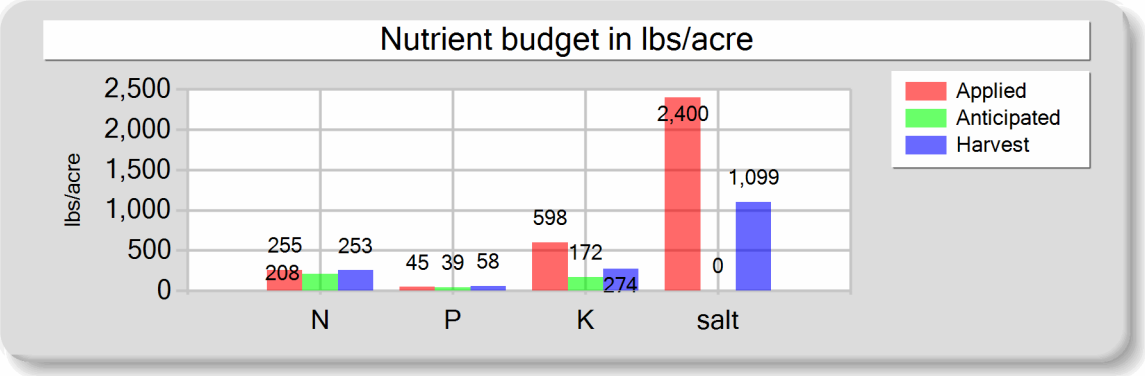
Fresh water applied	
38,664,000.00	*gallons*
1,423.86	*acre-inches*
39.55	*inches/acre*

Process wastewater applied	
4,530,120.00	*gallons*
166.83	*acre-inches*
4.63	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 04/18/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 04/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.37	45.21	597.63	2,400.48
Fresh water	22.05	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.42	45.21	597.63	2,400.48
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	253.05	58.20	273.93	1,098.96
Nutrient balance	2.37	-12.99	323.70	1,301.53
Applied to removed ratio	1.01	0.78	2.18	2.18

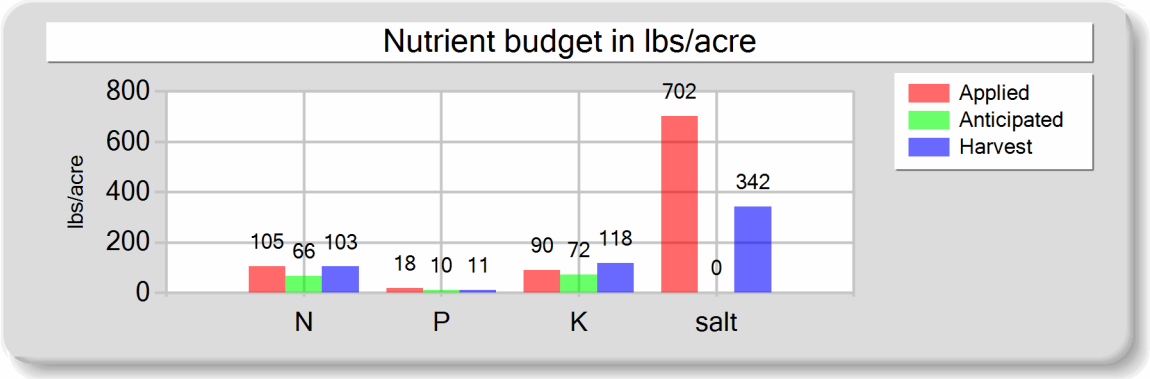
Fresh water applied	
52,488,000.00	*gallons*
1,932.95	*acre-inches*
33.91	*inches/acre*

Process wastewater applied	
4,247,760.00	*gallons*
156.43	*acre-inches*
2.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 08/14/2023: Sudangrass, silage

Field name: 6 Crop: Sudangrass, silage Plant date: 08/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	85.00	17.74	89.84	702.03
Fresh water	12.80	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	104.80	17.74	89.84	702.03
Anticipated crop nutrient removal	66.00	10.20	72.00	0.00
Actual crop nutrient removal	103.42	10.69	117.63	341.91
Nutrient balance	1.38	7.05	-27.79	360.13
Applied to removed ratio	1.01	1.66	0.76	2.05

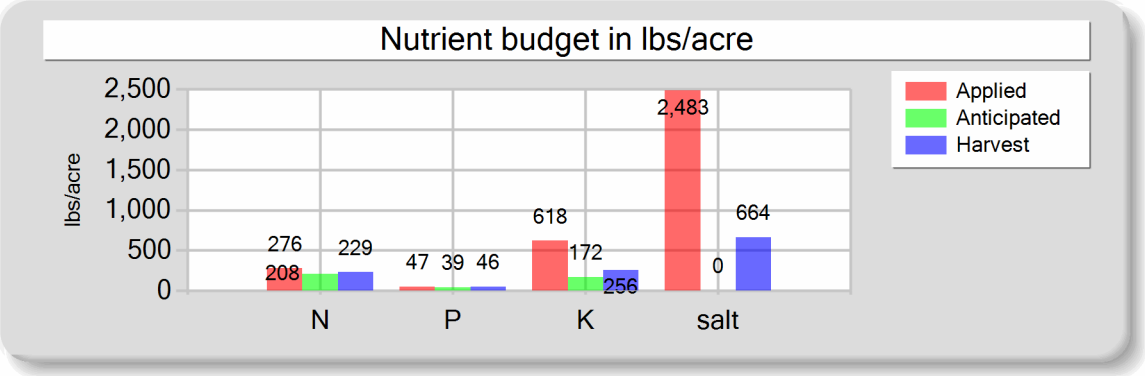
Fresh water applied	
30,456,000.00	*gallons*
1,121.59	*acre-inches*
19.68	*inches/acre*

Process wastewater applied	
6,480,000.00	*gallons*
238.64	*acre-inches*
4.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 04/20/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 04/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	234.18	46.77	618.27	2,483.37
Fresh water	27.46	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	275.64	46.77	618.27	2,483.37
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	228.82	46.48	256.41	663.51
Nutrient balance	46.82	0.29	361.86	1,819.86
Applied to removed ratio	1.20	1.01	2.41	3.74

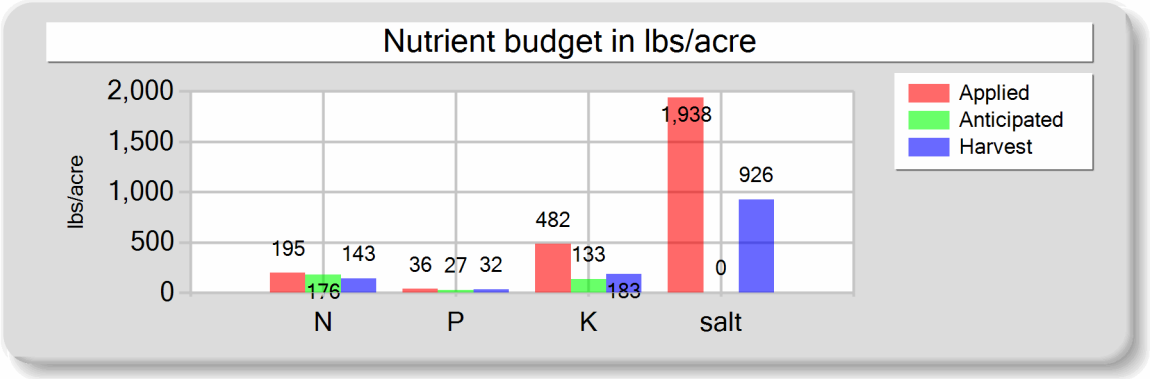
Fresh water applied	
14,904,000.00	*gallons*
548.86	*acre-inches*
42.22	*inches/acre*

Process wastewater applied	
1,002,240.00	*gallons*
36.91	*acre-inches*
2.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/02/2022: Wheat, silage, soft dough

Field name: 8 Crop: Wheat, silage, soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.74	36.50	482.46	1,937.86
Fresh water	5.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.49	36.50	482.46	1,937.86
Anticipated crop nutrient removal	176.00	27.20	132.80	0.00
Actual crop nutrient removal	142.73	32.13	182.58	926.16
Nutrient balance	52.76	4.37	299.88	1,011.69
Applied to removed ratio	1.37	1.14	2.64	2.09

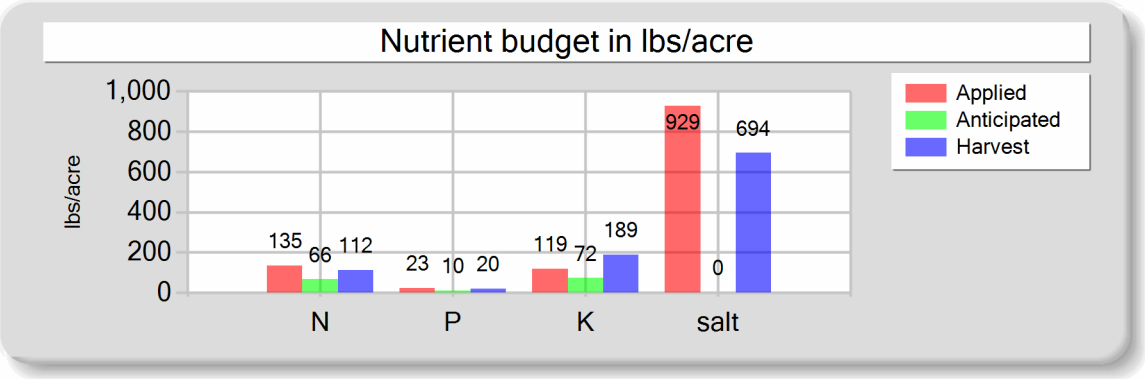
Fresh water applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
8.84	*inches/acre*

Process wastewater applied	
541,440.00	*gallons*
19.94	*acre-inches*
2.22	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 07/29/2023: Sudangrass, silage

Field name: 8 Crop: Sudangrass, silage Plant date: 07/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	112.46	23.47	118.86	928.77
Fresh water	15.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	134.98	23.47	118.86	928.77
Anticipated crop nutrient removal	66.00	10.20	72.00	0.00
Actual crop nutrient removal	112.39	19.63	188.60	693.68
Nutrient balance	22.58	3.84	-69.74	235.09
Applied to removed ratio	1.20	1.20	0.63	1.34

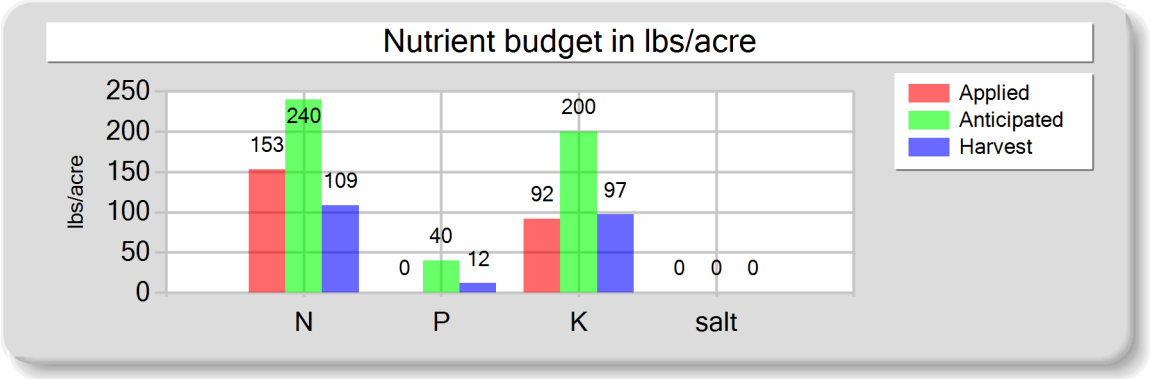
Fresh water applied	
5,832,000.00	*gallons*
214.77	*acre-inches*
23.86	*inches/acre*

Process wastewater applied	
1,353,600.00	*gallons*
49.85	*acre-inches*
5.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pistachios - 01/01/2022: Pistachios

Field name: Pistachios Crop: Pistachios Plant date: 01/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	115.00	0.00	92.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	23.86	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	152.86	0.00	92.00	0.00
Anticipated crop nutrient removal	240.00	40.00	200.00	0.00
Actual crop nutrient removal	108.62	11.64	96.98	0.00
Nutrient balance	44.24	-11.64	-4.98	0.00
Applied to removed ratio	1.41	0.00	0.95	0.00

Fresh water applied	
462,240,000.00	*gallons*
17,022.73	*acre-inches*
36.69	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Dry Manure

Sample and source description: Dry Manure

Sample date: 12/27/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 35.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,700.00	11,500.00	4,530.00							0.00
DL	100.00	100.00	100.00							0.01

Solid Manure

Sample and source description: Solid Manure

Sample date: 02/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,030.00	5,390.00							0.00
DL	100.00	100.00	100.00							0.01

Dry Manure

Sample and source description: Dry Manure

Sample date: 10/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	9,000.00	34,000.00							0.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

WWQ4

Sample and source description: WWQ4

Sample date: 12/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	722.00	437.00	0.00	0.00	88.40	643.00								6,190.00	4,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 12/27/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	722.00	437.00	0.00	0.00	88.40	643.00								6,190.00	4,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 04/05/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	563.00	403.00	0.00	0.00	40.60	383.00								6,310.00	4,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	364.00	300.00	0.00	0.00	72.70	961.00								5,890.00	3,860
DL	0.50	0.50	0.50	5.00	0.50	0.50								1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 09/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	89.60	58.80	0.00	0.00	18.70	94.70								1,460.00	740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

10990 Bear Creek

10990 Bear Creek

Sample description: 10990 Bear Creek

Sample date: 09/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

10990 Bear Creek

Sample description: 10990 Bear Creek

Sample date: 09/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								326.00	0
DL	0.50	0.50	0.50								1.00	10

D135

D135

D135

Sample description: D135

Sample date: 09/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

MID Canal

MID Canal

Sample description: MID Canal

Sample date: 09/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.87	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 04/12/2023: Corn, silage

1 - 04/12/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,060.00	760.00	5,550.00		5.15
DL	100.00	100.00	100.00		0.01

12 - 04/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,360.00	780.00	5,350.00		4.28
DL	100.00	100.00	100.00		0.01

12 - 08/15/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,880.00	900.00	9,440.00		10.60
DL	100.00	100.00	100.00		0.01

13 - 04/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,820.00	540.00	3,650.00		5.18
DL	100.00	100.00	100.00		0.01

13 - 08/16/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,020.00	1,440.00	14,900.00		10.90
DL	100.00	100.00	100.00		0.01

14 - 11/06/2022: Wheat Hay

Wheat

Sample and source description: Wheat

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,130.00	1,830.00	10,400.00		12.10
DL	100.00	100.00	100.00		0.01

14 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,760.00	5,000.00	4,220.00		7.30
DL	100.00	100.00	100.00		0.01

15 - 11/09/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,470.00	1,090.00	7,360.00		9.80
DL	100.00	100.00	100.00		0.01

15 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,310.00	1,160.00	4,380.00		6.50
DL	100.00	100.00	100.00		0.01

16 - 11/08/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,710.00	890.00	6,490.00		10.50
DL	100.00	100.00	100.00		0.01

16 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	1,130.00	6,450.00		6.50
DL	100.00	100.00	100.00		0.01

17 - 04/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	700.00	4,780.00		6.26
DL	100.00	100.00	100.00		0.01

18 - 11/03/2022: WheatHay

Wheat

Sample and source description: Wheat

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,090.00	1,550.00	9,700.00		10.60
DL	100.00	100.00	100.00		0.01

18 - 07/29/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,140.00	1,020.00	9,890.00		10.30
DL	100.00	100.00	100.00		0.01

2 - 04/13/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,590.00	670.00	6,000.00		5.68
DL	100.00	100.00	100.00		0.01

20 N - 04/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,270.00	930.00	5,310.00		5.80
DL	100.00	100.00	100.00		0.01

21 - 04/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,750.00	980.00	5,810.00		6.05
DL	100.00	100.00	100.00		0.01

24 E - 11/10/2022: Wheat, silage, soft dough

Forage

Sample and source description: Forage

Sample date: 05/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,610.00	1,070.00	6,410.00		9.50
DL	100.00	100.00	100.00		0.01

24 E - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,840.00	850.00	4,390.00		2.59
DL	100.00	100.00	100.00		0.01

25 - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,320.00	820.00	4,550.00		2.80
DL	100.00	100.00	100.00		0.01

26 - 11/11/2022: Wheat, silage, soft dough

Wheat

Sample and source description: Wheat

Sample date: 04/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,080.00	1,400.00	7,090.00		11.00
DL	100.00	100.00	100.00		0.01

26 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	580.00	4,830.00		6.96
DL	100.00	100.00	100.00		0.01

3 - 04/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	780.00	5,960.00		0.76
DL	100.00	100.00	100.00		0.01

4 - 04/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,080.00	690.00	5,120.00		4.20
DL	100.00	100.00	100.00		0.01

4 - 08/14/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,630.00	720.00	7,830.00		8.40
DL	100.00	100.00	100.00		0.01

5 - 04/17/2002: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	830.00	4,420.00		6.18
DL	100.00	100.00	100.00		0.01

6 - 04/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	920.00	4,330.00		5.17
DL	100.00	100.00	100.00		0.01

6 - 08/14/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,350.00	760.00	8,360.00		7.50
DL	100.00	100.00	100.00		0.01

7 - 04/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 08/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,480.00	910.00	5,020.00		3.68
DL	100.00	100.00	100.00		0.01

8 - 07/29/2023: Sudangrass, silage

Sudan

Sample and source description: Sudan

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,330.00	1,280.00	12,300.00		11.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

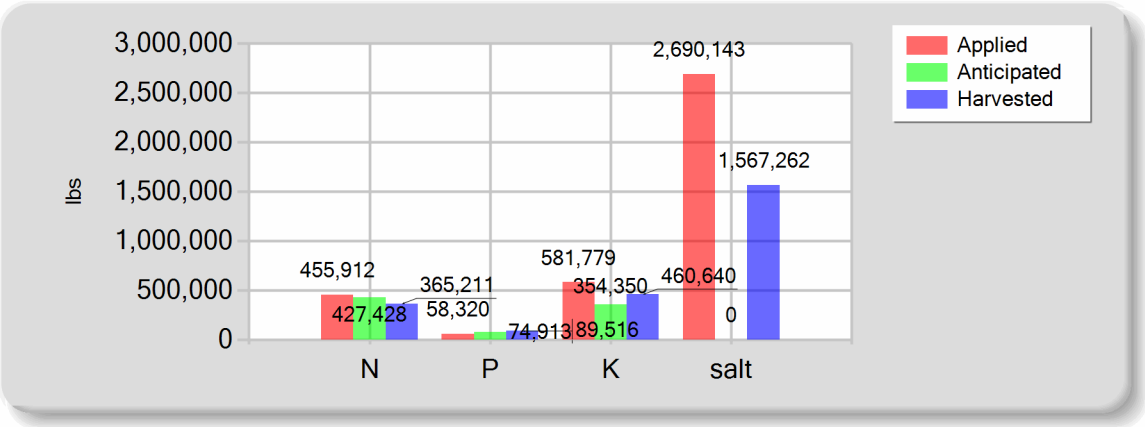
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

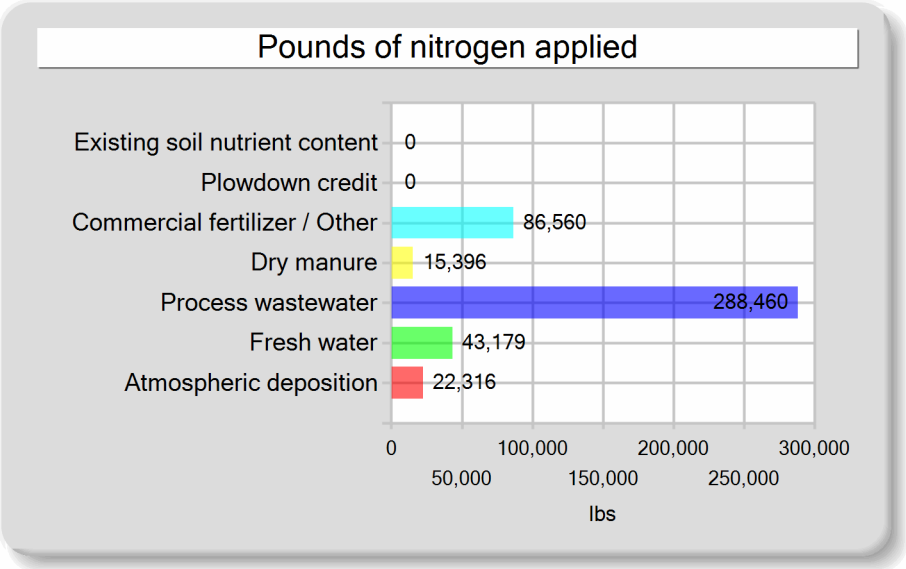
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

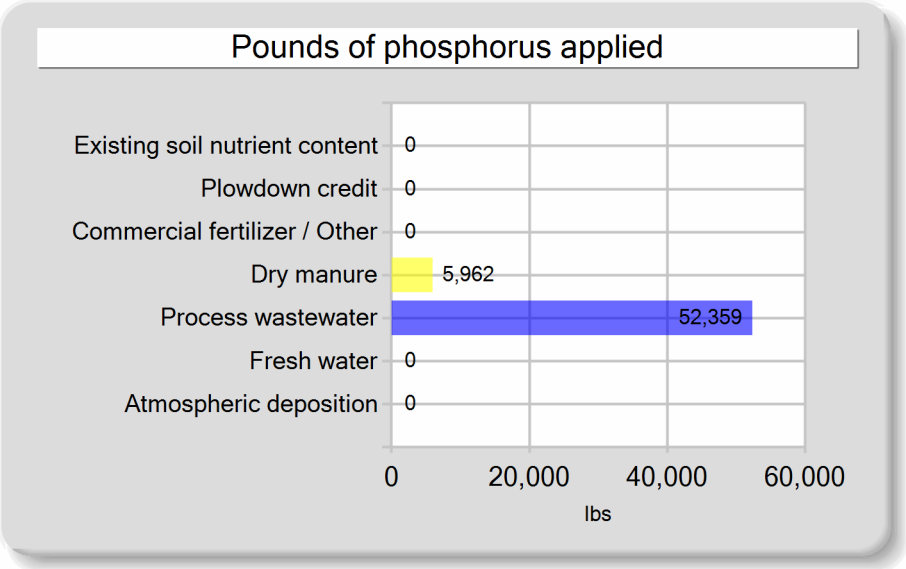
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	86,560.00	0.00	42,688.00	0.00
Dry manure	15,396.48	5,961.60	2,348.35	0.00
Process wastewater	288,459.95	52,358.59	536,743.09	2,690,142.83
Fresh water	43,179.44	0.00	0.00	0.00
Atmospheric deposition	22,316.00	0.00	0.00	0.00
Total nutrients applied	455,911.87	58,320.19	581,779.45	2,690,142.83
Anticipated crop nutrient removal	427,428.00	74,912.60	354,349.60	0.00
Actual crop nutrient removal	365,211.46	89,515.98	460,639.88	1,567,261.93
Nutrient balance	90,700.41	-31,195.79	121,139.57	1,122,880.90
Applied to removed ratio	1.25	0.65	1.26	1.72

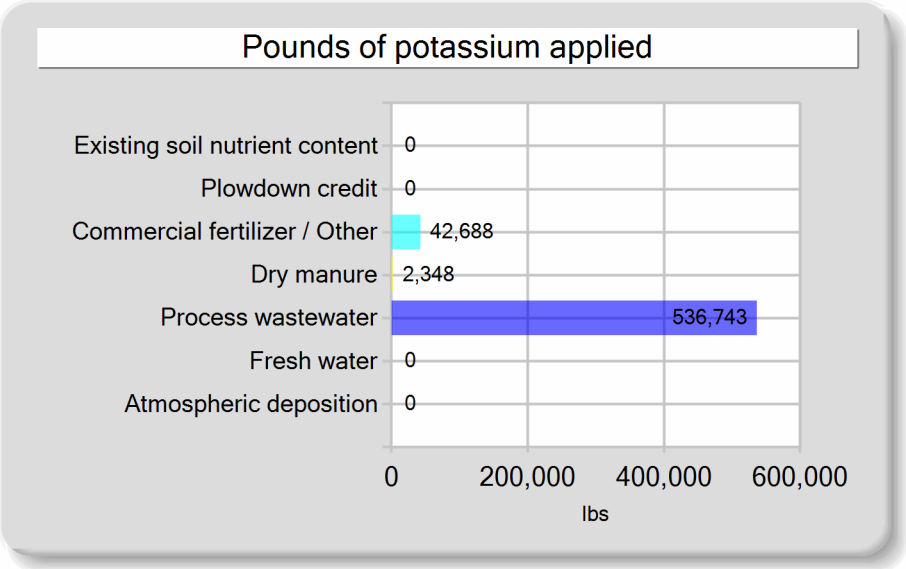
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

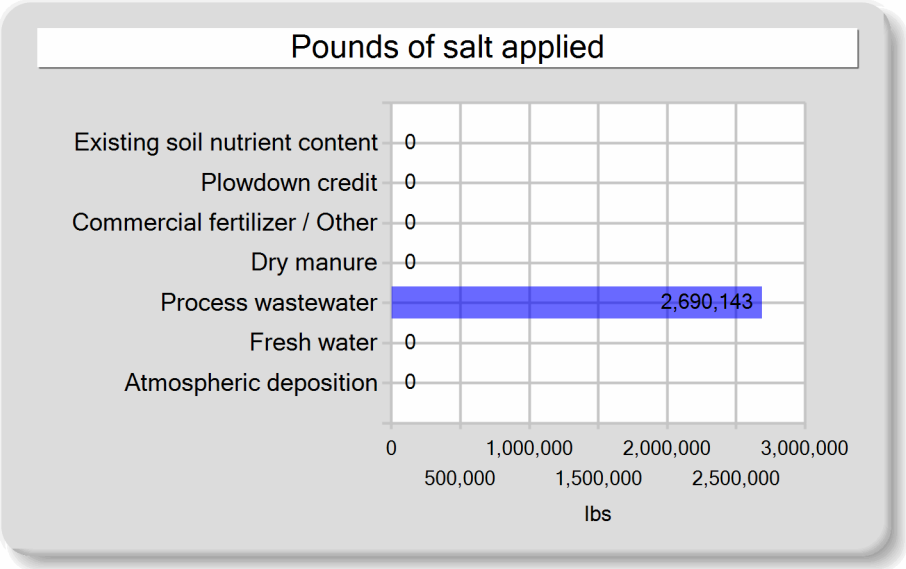


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Edward L Hoekstra (May 3, 2024 07:17 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Edward Hoekstra	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 3, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Thomas

Name of Dairy Facility: Hillcrest Dairy LLC

Facility Address:

1901 N Hayden RD	Le Grand	Merced	95333
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Thomas (209) 535-8591
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons Custom Spreading

Address of Hauling Company/Person:

P.O. Box 1261	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Silva & Sons (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Press Dairy (209) 535-8591
Name Phone Number

10310 Reese RD	Planada	CA	95365
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

0053-0080-0012-0000 Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/06/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,000.00 tons

Manure Solids Content: 66.6 %

Method used to determine amount of manure:

Scale Weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Edward L Hoekstra (May 3, 2024 07:17 PDT)

May 3, 2024

Operator Signature — Date

Frank Silva

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Thomas

Name of Dairy Facility: Hillcrest Dairy LLC

Facility Address:

1901 N Hayden RD	Le Grand	Merced	95333
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Thomas (209) 535-8591
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons Custom Spreading

Address of Hauling Company/Person:

P.O. Box 1261	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Silva & Sons (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Press Dairy (209) 535-8591
Name Phone Number

10310 Reese RD	Planada	CA	95365
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

0053-0080-0012-0000 Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/06/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 500.00 tons

Manure Solids Content: 66.6 %

Method used to determine amount of manure:

Scale Weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Edward L Hoekstra (May 3, 2024 07:17 PDT)

May 3, 2024

Operator Signature — Date

Frank Silva

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Thomas

Name of Dairy Facility: Hillcrest Dairy LLC

Facility Address:

1901 N Hayden RD	Le Grand	Merced	95333
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Thomas (209) 535-8591
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons Custom Spreading

Address of Hauling Company/Person:

P.O. Box 1261	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Silva & Sons (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Press Dairy (209) 535-8591
Name Phone Number

10310 Reese RD	Planada	CA	95365
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

0053-0080-0028-0000 Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/06/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 750.00 tons

Manure Solids Content: 66.6 %

Method used to determine amount of manure:

Scale Weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Edward L Hoekstra (May 3, 2024 07:17 PDT)

May 3, 2024

Operator Signature | Date

Frank Silva

Hauler Signature | Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Thomas

Name of Dairy Facility: Hillcrest Dairy LLC

Facility Address:

1901 N Hayden RD	Le Grand	Merced	95333
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Thomas (209) 535-8591
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Silva & Sons Custom Spreading

Address of Hauling Company/Person:

P.O. Box 1261	Hilmar	CA	95324
Number and Street	City	State	Zip Code

Contact Person: Silva & Sons (209) 632-8202
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Press Dairy (209) 535-8591
Name Phone Number

10310 Reese RD	Planada	CA	95365
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

Address City Zip Code

Street and nearest cross street (if no address) County

0053-0080-0028-0000 Merced
Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 09/06/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 750.00 tons

Manure Solids Content: 66.6 %

Method used to determine amount of manure:

Scale Weights

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Edward L Hoekstra (May 3, 2024 07:17 PDT)

May 3, 2024

Operator Signature | Date

Frank Silva

Hauler Signature | Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Hillcrest Dairy, LLC
1901 North Hayden Rd
Le Grand, CA 95333

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/12/23 9:55 AM
SUBMITTED BY: John
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 19.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
10990 Bear Creek	T3255009B	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.235	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	48.7	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	98.2 (T3255009B)	96.4 (T3255009B)	1.85
Electrical Conductivity (EC)	97.1			0.590
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)	1.02

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Hillcrest Dairy, LLC
1901 North Hayden Rd
Le Grand, CA 95333

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/12/23 9:55 AM
SUBMITTED BY: John
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 19.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
D135	T3255009C	Electrical Conductivity (EC)	EPA 120.1 and	0.412	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	264	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	98.2 (T3255009B)	96.4 (T3255009B)	1.85
Electrical Conductivity (EC)	97.1			0.590
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)	1.02

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Compliance Analysis Report

Customer
Hillcrest Dairy, LLC
1901 North Hayden Rd
Le Grand, CA 95333

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/12/23 9:55 AM
SUBMITTED BY: John
ANALYZED DATE: 9/12/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
10090 Bear Creek	T3255009D	Electrical Conductivity (EC)	EPA 120.1 and	0.326	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	209	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	98.2 (T3255009B)	96.4 (T3255009B)	1.85
Electrical Conductivity (EC)	97.1			0.590
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)	1.02

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Compliance Analysis Report

Customer
Hillcrest Dairy, LLC
1901 North Hayden Rd
Le Grand, CA 95333

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/12/23 9:55 AM
SUBMITTED BY: John
ANALYZED DATE: 9/12/23

Irrigation/Ground Water (H6)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 19.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
MID	T3255009E	Total Dissolved Solids (TDS)	SM 2540 C-1997 and	< 10.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	2.87	mg/L
		Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Electrical Conductivity (EC)	97.1			1.70 (T3255004A)	1.69 (Dup)	0.590
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)			1.02
Total Dissolved Solids (TDS)	93.0			1520 (T3255018A)	1550 (Dup)	1.96
Total Kjeldahl Nitrogen (TKN)	99.1			2.80 (T3240027F)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Hillcrest Dairy
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F+R Services
Sampled By: John Hue
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	9-12-23	7:30 AM	3rd QTR Lagoon		21.2°			X			
2			10990 Bear Creek		19.8°		X				
3			D135		19.6°		X				
4			10090 Bear Creek		21.0°		X				
5			Mid		19.8°		~~X~~ AM				H6
6			~~D199~~				~~X~~				
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]					
2nd	[signature]	DENELE ANALYTICAL, INC.	9-12-23 9:55			T3255009

Required to be submitted to any agency? (YES / NO)

Binder1

Final Audit Report 2024-05-03

Created:	2024-04-30
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAI_zX1S-ztY5LfODNX027usReiNdO3Lbi

"Binder1" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-04-30 - 5:47:09 PM GMT

Document emailed to elhoekstra@yahoo.com for signature
2024-04-30 - 5:51:13 PM GMT

Email viewed by elhoekstra@yahoo.com
2024-04-30 - 10:43:43 PM GMT

Signer elhoekstra@yahoo.com entered name at signing as Edward L Hoekstra
2024-05-03 - 2:17:14 PM GMT

Document e-signed by Edward L Hoekstra (elhoekstra@yahoo.com)
Signature Date: 2024-05-03 - 2:17:16 PM GMT - Time Source: server

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